UE26CY151A · ENGINEERING CHEMISTRY

SEMESTER 1 · UNIT 1

06

QUESTION BANK

Every Unit 1 question — solved, then drilled

All Unit 1 questions from the past ESA papers (Question 1 of each paper, 2023–2025), each worked out step by step, followed by several practice variants of every question type so you can see the same idea from different angles.

38
solved past questions

122
practice questions with answers

11
theory model answers

How to use	Read the recap → study the solved past questions → attempt Section B without looking → check against the answer key.
Papers	12 distinct ESA papers (UE22, UE23, UE24 syllabi); duplicate papers removed.
Flags	NOT ON CURRENT SLIDES marks topics from older syllabi (Fe–C, CO_2 diagram, Beer–Lambert).

Numerical answers are computed by script from each question's own constants, so small differences from a printed key come from rounding of constants. · 13 October 2026

Contents

Q1 Rotational (microwave) spectroscopy — 9 past · 26 practice

Q2 Vibrational (IR) spectroscopy – harmonic oscillator — 8 past · 19 practice

Q3 Anharmonic oscillator and overtones — 5 past · 18 practice

Q4 Born–Oppenheimer, electronic spectra, Franck–Condon and Raman — 6 past · 14 practice

Q5 Phase rule, water and Pb–Ag systems — 9 past · 35 practice

Q6 Computational chemistry — 1 past · 10 practice

WHAT THE EXAMINERS ASK — QUESTION TYPES

Question type	Chapter	Times asked	Practice variants here
Rigid rotor: derive $I = \mu r_o^2$	Q1	4	1
Rigid rotor: bond length from first line (HCl, HI, HBr)	Q1	4	6
Rigid rotor: I and spacing from B; level diagram J ≤ 5/6 and spectrum	Q1	4	5
Rigid rotor: J→J+1 line; first line from I or r_o	Q1	2	5
SHO: $\bar{\nu}$ and zero-point energy from k (HF ×2, NO, CO)	Q2	5	6
SHO: k from $\bar{\nu}$ (CO ×3, HBr)	Q2	4	5
SHO: why one line; CO_2 and HI IR activity	Q2	6	4
Anharmonic: x_e, ω_e, overtone calculations	Q3	3	12
Anharmonic: selection rule, B–O diagram, SHO failure	Q3	3	4
Franck–Condon statement, three cases, spectra	Q4	5	5
Continuum / B–O / Raman / Stokes / Beer–Lambert	Q4	5	6
Phase rule: derivation	Q5	3	5
Phase rule: count C, P, F	Q5	3	12
Water diagram, F values, metastable curve, Clausius–Clapeyron	Q5	4	9
Reduced phase rule and Pb–Ag diagram, cooling curves	Q5	4	9
Computational chemistry: definition, applications, methods	Q6	1	10

Not Unit 1 any more. Conductometric titration, colorimetry and α-Al_2O_3 synthesis appear in older Q1s but belong to other units/labs; they are left out. Papers 1, 4 and 6 are the same paper. Paper "Jan · UE23A Set 2" has Q1.b as an image, which was read directly.

CHAPTER 1 · TOPICS T4

Rotational (microwave) spectroscopy

9 solved past questions | 26 practice questions | 0 theory answers

The rigid-rotor chapter is the most heavily examined part of Unit 1: a bond-length calculation, the moment-of-inertia derivation and a level/spectrum sketch appear almost every sitting.

RECAP · FORMULAS YOU NEED

$$\varepsilon_J = BJ(J+1)\ \text{cm}^{-1}$$

$$B = \frac{h}{8\pi^2 Ic}$$

$$I = \mu r_0^2,\quad \mu = \frac{m_1 m_2}{m_1 + m_2}$$

$$\Delta J = \pm 1,\quad \text{line}(J \to J+1) = 2B(J+1)$$

$$\text{first line} = 2B,\quad \text{spacing} = 2B$$

METHOD

1. Read the first line (or the spacing): $B = \text{line}/2$.
2. $I = h/(8\pi^2 cB)$ with c in cm s^{-1}.
3. μ in kg: amu × 1.66×10^{-27}.
4. $r_0 = \sqrt{I/\mu}$, then convert m → Å.

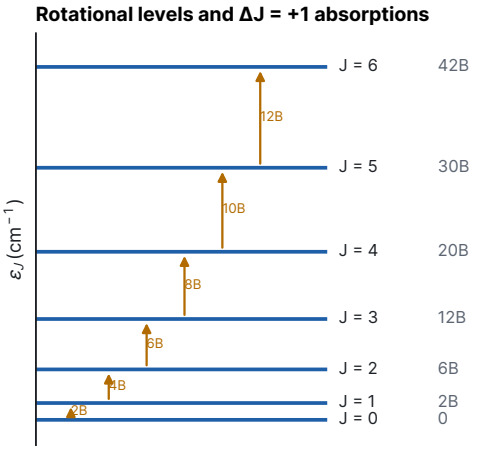


Fig. G9

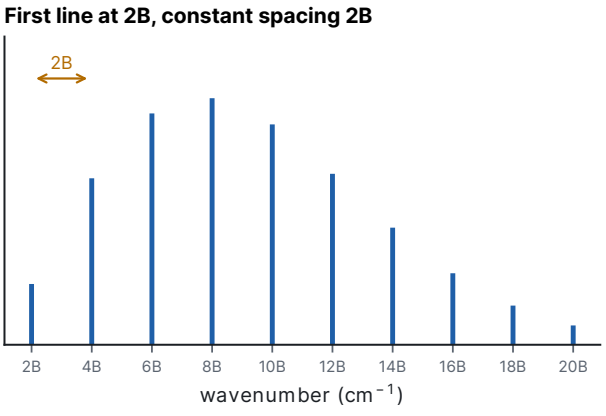


Fig. G10

Q1-D · DERIVATION: I = MR$_0^2$ (ASKED ×4)

1. Let masses m_1, m_2 be at distances r_1, r_2 from the centre of mass C, with $r_0 = r_1 + r_2$.
2. Balance about C: $m_1 r_1 = m_2 r_2$. Substitute $r_2 = r_0 - r_1$: $m_1 r_1 = m_2(r_0 - r_1) \Rightarrow r_1 = \dfrac{m_2 r_0}{m_1 + m_2}$ and $r_2 = \dfrac{m_1 r_0}{m_1 + m_2}$.
3. $I = m_1 r_1^2 + m_2 r_2^2 = \dfrac{m_1 m_2^2 r_0^2 + m_2 m_1^2 r_0^2}{(m_1 + m_2)^2} = \dfrac{m_1 m_2 (m_1 + m_2) r_0^2}{(m_1 + m_2)^2}$.
4. $\therefore\ I = \dfrac{m_1 m_2}{m_1 + m_2} r_0^2 = \mu r_0^2$. (Also the energy: $E_r = \frac{1}{2} I\omega^2 = L^2/2I$, giving $E_J = \dfrac{h^2 J(J+1)}{8\pi^2 I}$.)

A · PAST ESA QUESTIONS, FULLY SOLVED

Q1.1 For a rigid-rotor heteronuclear diatomic molecule AX the first line in the pure rotational spectrum is at 21.18 cm^{-1}. Derive an expression for the moment of inertia and calculate the moment of inertia and bond length of AX. (A = 1.008 amu, X = 35.45 amu; $c = 3 \times 10^{10}$ cm s^{-1}, $\pi = 3.14$, $h = 6.627 \times 10^{-34}$ J s, 1 amu $= 1.66 \times 10^{-27}$ kg)

Jun-24 · UE23 · Set 2 · 6 marks

1. **Derivation:** see Q1-D above (I = μr$_0^2$).
2. First line at $2B$, so $B = \dfrac{21.18}{2} = 10.59$ cm^{-1}.
3. $I = \dfrac{h}{8\pi^2 cB} = \dfrac{6.63 \times 10^{-34}}{8(3.14)^2(3 \times 10^{10})(10.59)} = 2.64 \times 10^{-47}$ kg m^2 (used $c = 3 \times 10^{10}$ cm s^{-1} because B is in cm^{-1})
4. $\mu = \dfrac{m_1 m_2}{m_1 + m_2} = \dfrac{1.008 \times 35.45}{36.46} = 0.9801$ amu $= 1.63 \times 10^{-27}$ kg.
5. $r_0 = \sqrt{I/\mu} = \sqrt{\dfrac{2.64 \times 10^{-47}}{1.63 \times 10^{-27}}} = 1.27 \times 10^{-10}$ m $= 1.275$ Å.

ANSWER $I = 2.64 \times 10^{-47}$ **kg m²**, $r_0 = 1.27$ **Å**

Q1.2 The first line in the pure rotational spectrum of HCl appears at 21.18 cm^{-1}. Calculate the bond length. (H = 1.008, Cl = 35.45 amu; 1 amu $= 1.66 \times 10^{-27}$ kg, $c = 3 \times 10^8$ m s^{-1}, $h = 6.6 \times 10^{-34}$ J s)

Dec-23 · UE22 · 4 marks

1. Because c is given in m s^{-1}, convert B to m^{-1} (×100) or use $c = 3 \times 10^{10}$ cm s^{-1} — same result.
2. First line at $2B$, so $B = \dfrac{21.18}{2} = 10.59$ cm^{-1}.
3. $I = \dfrac{h}{8\pi^2 cB} = \dfrac{6.60 \times 10^{-34}}{8(3.14)^2(3 \times 10^{10})(10.59)} = 2.63 \times 10^{-47}$ kg m^2 (used $c = 3 \times 10^{10}$ cm s^{-1} because B is in cm^{-1})
4. $\mu = \dfrac{m_1 m_2}{m_1 + m_2} = \dfrac{1.008 \times 35.45}{36.46} = 0.9801$ amu $= 1.63 \times 10^{-27}$ kg.
5. $r_0 = \sqrt{I/\mu} = \sqrt{\dfrac{2.63 \times 10^{-47}}{1.63 \times 10^{-27}}} = 1.27 \times 10^{-10}$ m $= 1.272$ Å.

ANSWER $r_0 \approx 1.27$ **Å**

Q1.3 For HBr, the rotational spectrum shows the first line at 17.19 cm^{-1}. Calculate the bond length (rigid rotor). (H = 1.0, Br = 79.9 amu; $h = 6.627 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1})

Jul-23 · UE22 · A · 4 marks

1. First line at $2B$, so $B = \dfrac{17.19}{2} = 8.595\ \text{cm}^{-1}$.
2. $I = \dfrac{h}{8\pi^2 cB} = \dfrac{6.63 \times 10^{-34}}{8(3.14)^2(3 \times 10^{10})(8.595)} = 3.26 \times 10^{-47}\ \text{kg m}^2$ (used $c = 3 \times 10^{10}$ cm s^{-1} because B is in cm^{-1})
3. $\mu = \dfrac{m_1 m_2}{m_1 + m_2} = \dfrac{1.0 \times 79.9}{80.9} = 0.9876\ \text{amu} = 1.64 \times 10^{-27}\ \text{kg}$.
4. $r_0 = \sqrt{I/\mu} = \sqrt{\dfrac{3.26 \times 10^{-47}}{1.64 \times 10^{-27}}} = 1.41 \times 10^{-10}\ \text{m} = 1.409\ \text{Å}$.

ANSWER $r_0 \approx 1.41$ **Å**

Q1.4 The first spectral line in the rotational spectrum of HI appears at 13.022 cm^{-1}. Calculate the bond length. (H = 1.0, I = 126.9 amu; $c = 3 \times 10^8$ m s^{-1}, 1 amu $= 1.66 \times 10^{-27}$ kg, $\pi = 3.14$, $h = 6.627 \times 10^{-34}$ J s)

Jan-25 · UE24 · Set 1 · 3 marks

1. First line at $2B$, so $B = \dfrac{13.022}{2} = 6.511\ \text{cm}^{-1}$.
2. $I = \dfrac{h}{8\pi^2 cB} = \dfrac{6.63 \times 10^{-34}}{8(3.14)^2(3 \times 10^{10})(6.511)} = 4.30 \times 10^{-47}\ \text{kg m}^2$ (used $c = 3 \times 10^{10}$ cm s^{-1} because B is in cm^{-1})
3. $\mu = \dfrac{m_1 m_2}{m_1 + m_2} = \dfrac{1.0 \times 126.9}{127.9} = 0.9922\ \text{amu} = 1.65 \times 10^{-27}\ \text{kg}$.
4. $r_0 = \sqrt{I/\mu} = \sqrt{\dfrac{4.30 \times 10^{-47}}{1.65 \times 10^{-27}}} = 1.62 \times 10^{-10}\ \text{m} = 1.616\ \text{Å}$.

ANSWER $I \approx 4.30 \times 10^{-47}$ **kg m²**, $r_0 \approx 1.62$ **Å**

Q1.5 For a rigid-rotor diatomic molecule: (i) draw the energy-level diagram up to $J = 5$; (ii) draw the rotational spectrum; (iii) calculate the moment of inertia and the distance between two spectral lines if $B = 10.56\ \text{cm}^{-1}$. ($c = 3 \times 10^8$ m s^{-1}, $\pi = 3.14$, $h = 6.625 \times 10^{-34}$ J s)

Jun-24 · UE23 · Set 1 · 6 marks

1. (i) Levels $\varepsilon_J = BJ(J+1)$: 0, 2B, 6B, 12B, 20B, 30B for J = 0…5 — gaps grow 2B, 4B, 6B, 8B, 10B (**Fig. G9**).
2. (ii) Allowed lines $J \to J+1$ at 2B, 4B, 6B, 8B, 10B… equally spaced by 2B; intensity rises then falls (**Fig. G10**).
3. (iii) $I = \dfrac{h}{8\pi^2 cB} = \dfrac{6.625 \times 10^{-34}}{8(3.14)^2(3 \times 10^{10})(10.56)} = 2.65 \times 10^{-47}\ \text{kg m}^2$.
4. Spacing between adjacent lines $= 2B = 2(10.56) = 21.12\ \text{cm}^{-1}$.

ANSWER $I = 2.65 \times 10^{-47}$ **kg m², spacing** $= 21.12\ \text{cm}^{-1}$

Q1.6 For a rotating diatomic molecule: (i) derive an expression for the moment of inertia I; (ii) if $B = 10\ \text{cm}^{-1}$, give the rotational energy levels for $J = 0, 1, 2, 3$ in cm^{-1} and draw the energy-level diagram up to $J = 3$.

Jan · UE23A · Set 2 | 6 marks

1. (i) See derivation Q1-D.
2. (ii) $\varepsilon_J = 10\,J(J+1)$: $J=0 \to 0$; $J=1 \to 20$; $J=2 \to 60$; $J=3 \to 120\ \text{cm}^{-1}$. Draw four horizontal lines at these heights (gaps 20, 40, 60) with $\Delta J = +1$ arrows.

ANSWER **Levels: 0, 20, 60, 120 cm^{-1}**

Q1.7 For a rigid-rotor diatomic molecule draw the energy-level diagram up to $J = 6$.

Dec-24 · UE22 | 2 marks

1. $\varepsilon_J = BJ(J+1)$: 0, 2B, 6B, 12B, 20B, 30B, 42B — see **Fig. G9** (it is drawn to J = 6). Label J on each level and show gaps increasing 2B, 4B, … 12B.

ANSWER **Levels 0, 2B, 6B, 12B, 20B, 30B, 42B**

Q1.8 (ii) If the rotational transition $J = 0 \to 1$ for HCl occurs at $20.68\ \text{cm}^{-1}$, find the energy required for the $J = 20 \to 21$ transition in cm^{-1}. (iii) If the moment of inertia of HBr is 3.26×10^{-47} kg m², at what wavenumber (in m^{-1}) does the first line appear? ($h = 6.62 \times 10^{-34}$ J s, $c = 3 \times 10^{8}$ m s^{-1})

Jul-23 · UE22 · B | 6 marks

1. (ii) $J = 0 \to 1$ gives $2B = 20.68$, so $B = 10.34\ \text{cm}^{-1}$. $\Delta\varepsilon(J \to J+1) = 2B(J+1)$; for $J = 20$: $2B \times 21 = 21 \times 20.68 = \mathbf{434.28}\ \text{cm}^{-1}$.
2. (iii) $B = \dfrac{h}{8\pi^2 Ic} = \dfrac{6.62 \times 10^{-34}}{8\pi^2(3.26 \times 10^{-47})(3 \times 10^{8})} = 857.3\ \text{m}^{-1}$ (c in m s^{-1} gives m^{-1} directly).
3. First line $= 2B = 1715\ \text{m}^{-1} = 17.15\ \text{cm}^{-1}$.

ANSWER **(ii) 434.28 cm^{-1} · (iii) ≈ 1715 m^{-1}**

Q1.9 For HBr, with bond length 1.404 Å, at what energy (in m^{-1}) does the first line appear in the rotational spectrum? (H = 1.008, Br = 79.9 amu; 1 amu $= 1.66 \times 10^{-27}$ kg, $c = 3 \times 10^{8}$ m s^{-1}, $h = 6.6 \times 10^{-34}$ J s)

Jan-25 · UE24 · Set 2 | 6 marks

1. $\mu = \dfrac{1.008 \times 79.9}{80.908}$ amu $= 0.9954$ amu $= 1.65 \times 10^{-27}$ kg.
2. $I = \mu r_0^2 = 1.65 \times 10^{-27} \times (1.404 \times 10^{-10})^2 = 3.26 \times 10^{-47}$ kg m².
3. $B = \dfrac{h}{8\pi^2 Ic} = 855.4\ \text{m}^{-1}$.
4. First line $= 2B = 1711\ \text{m}^{-1}$ ($\approx 17.11\ \text{cm}^{-1}$).

ANSWER **First line ≈ 1711 m^{-1}**

B · PRACTICE QUESTIONS (ANSWERS IN THE KEY AT THE END OF THIS CHAPTER)

Set 1 · Bond length from the first line

Same pattern as the exam. Do all six — the method never changes.

Q1.P1.1 The first line in the rotational spectrum of HF is at 41.9 cm^{-1}. Calculate B, I and the bond length. (H = 1.008, F = 18.998 amu; $h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1}, 1 amu $= 1.66 \times 10^{-27}$ kg)

Q1.P1.2 The first line in the rotational spectrum of CO is at 3.86 cm^{-1}. Calculate B, I and the bond length. (C = 12.0, O = 15.995 amu; $h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1}, 1 amu $= 1.66 \times 10^{-27}$ kg)

Q1.P1.3 The first line in the rotational spectrum of HBr is at 17.19 cm^{-1}. Calculate B, I and the bond length. (H = 1.008, Br = 79.904 amu; $h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1}, 1 amu $= 1.66 \times 10^{-27}$ kg)

Q1.P1.4 The first line in the rotational spectrum of HI is at 13.0 cm^{-1}. Calculate B, I and the bond length. (H = 1.008, I = 126.9 amu; $h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1}, 1 amu $= 1.66 \times 10^{-27}$ kg)

Q1.P1.5 The first line in the rotational spectrum of CS is at 1.632 cm^{-1}. Calculate B, I and the bond length. (C = 12.0, S = 31.97 amu; $h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1}, 1 amu $= 1.66 \times 10^{-27}$ kg)

Q1.P1.6 The first line in the rotational spectrum of HCl is at 21.18 cm^{-1}. Calculate B, I and the bond length. (H = 1.008, Cl = 35.45 amu; $h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1}, 1 amu $= 1.66 \times 10^{-27}$ kg)

Set 2 · Moment of inertia and line spacing from B

Watch the units: B in cm^{-1} pairs with c in cm s^{-1}.

Q1.P2.1 A rigid rotor has $B = 10.56$ cm^{-1}. Calculate its moment of inertia, the spacing between adjacent lines, and the positions of the first three lines. ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s^{-1})

Q1.P2.2 A rigid rotor has $B = 5.0$ cm^{-1}. Calculate its moment of inertia, the spacing between adjacent lines, and the positions of the first three lines. ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s^{-1})

Q1.P2.3 A rigid rotor has $B = 1.93$ cm^{-1}. Calculate its moment of inertia, the spacing between adjacent lines, and the positions of the first three lines. ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s^{-1})

Q1.P2.4 A rigid rotor has $B = 8.5\ \text{cm}^{-1}$. Calculate its moment of inertia, the spacing between adjacent lines, and the positions of the first three lines. ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s $^{-1}$)

Q1.P2.5 A rigid rotor has $B = 0.39\ \text{cm}^{-1}$. Calculate its moment of inertia, the spacing between adjacent lines, and the positions of the first three lines. ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s $^{-1}$)

Set 3 · Position of a high-J line

The trick: first line = 2B, so the J→J+1 line is simply (J+1) × first line.

Q1.P3.1 For HCl the $J = 0 \rightarrow 1$ line is at 20.68 cm^{-1}. Find the line for $J = 20 \rightarrow 21$ and the wavenumber gap between it and the previous line.

Q1.P3.2 For HCl the $J = 0 \rightarrow 1$ line is at 21.18 cm^{-1}. Find the line for $J = 9 \rightarrow 10$ and the wavenumber gap between it and the previous line.

Q1.P3.3 For HBr the $J = 0 \rightarrow 1$ line is at 17.19 cm^{-1}. Find the line for $J = 14 \rightarrow 15$ and the wavenumber gap between it and the previous line.

Q1.P3.4 For HI the $J = 0 \rightarrow 1$ line is at 13.022 cm^{-1}. Find the line for $J = 4 \rightarrow 5$ and the wavenumber gap between it and the previous line.

Q1.P3.5 For CO the $J = 0 \rightarrow 1$ line is at 3.86 cm^{-1}. Find the line for $J = 29 \rightarrow 30$ and the wavenumber gap between it and the previous line.

Set 4 · The reverse problem: from I or r_0 to the first line

Exam variants (HBr I = 3.26×10 $^{-47}$; HBr r_0 = 1.404 Å) are in this family.

Q1.P4.1 The moment of inertia of HBr is 3.26×10^{-47} kg m². At what wavenumber (cm^{-1}) does the first rotational line appear? ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s^{-1})

Q1.P4.2 The moment of inertia of HCl is 2.64×10^{-47} kg m². At what wavenumber (cm^{-1}) does the first rotational line appear? ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s^{-1})

Q1.P4.3 The bond length of HI is 1.61 Å. At what wavenumber (cm^{-1}) does the first rotational line appear? (H = 1.008, I = 126.9 amu) ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s^{-1})

Q1.P4.4 The bond length of CO is 1.128 Å. At what wavenumber (cm^{-1}) does the first rotational line appear? (C = 12.0, O = 15.995 amu) ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s^{-1})

Q1.P4.5 The bond length of HF is 0.917 Å. At what wavenumber (cm^{-1}) does the first rotational line appear? (H = 1.008, F = 18.998 amu) ($h = 6.626 \times 10^{-34}$, $c = 3 \times 10^{10}$ cm s^{-1})

Set 5 · Concept twists

Same ideas, asked differently.

Q1.P5.1 DCl has the same bond length as HCl (D = 2.014 amu). If HCl's first line is at 21.18 cm^{-1}, predict DCl's first line. Is it higher or lower? Why?

Q1.P5.2 A rotational spectrum shows lines at 42.4, 63.6 and 84.8 cm^{-1} with no line at 21.2 cm^{-1} shown. Deduce B and the J values involved.

Q1.P5.3 Which of these show a pure rotational spectrum: H_2, HCl, CO_2, CO, N_2, HF, Cl_2, H_2O? Give the rule.

Q1.P5.4 Why does the intensity of rotational lines first rise and then fall with J?

Q1.P5.5 How many lines appear between 0 and 100 cm^{-1} for $B = 10.56$ cm^{-1}?

C · ANSWER KEY

Set 1 · Bond length from the first line

Q1.P1.1 1. $B = 41.9/2 = 20.95$ cm^{-1}; $I = h/8\pi^2 cB = 1.34 \times 10^{-47}$ kg m^2; $\mu = 0.9572$ amu.

ANSWER $B = 20.95$ **cm**$^{-1}$, $I = 1.34 \times 10^{-47}$ **kg m²**, $r_0 = 0.917$ **Å**

Q1.P1.2 1. $B = 3.86/2 = 1.93$ cm^{-1}; $I = h/8\pi^2 cB = 1.45 \times 10^{-46}$ kg m^2; $\mu = 6.856$ amu.

ANSWER $B = 1.93$ **cm**$^{-1}$, $I = 1.45 \times 10^{-46}$ **kg m²**, $r_0 = 1.128$ **Å**

Q1.P1.3 1. $B = 17.19/2 = 8.595$ cm^{-1}; $I = h/8\pi^2 cB = 3.25 \times 10^{-47}$ kg m^2; $\mu = 0.9954$ amu.

ANSWER $B = 8.595$ **cm**$^{-1}$, $I = 3.25 \times 10^{-47}$ **kg m²**, $r_0 = 1.403$ **Å**

Q1.P1.4 1. $B = 13.0/2 = 6.5$ cm^{-1}; $I = h/8\pi^2 cB = 4.30 \times 10^{-47}$ kg m^2; $\mu = 1$ amu.

ANSWER $B = 6.5$ **cm**$^{-1}$, $I = 4.30 \times 10^{-47}$ **kg m²**, $r_0 = 1.610$ **Å**

Q1.P1.5 1. $B = 1.632/2 = 0.816$ cm^{-1}; $I = h/8\pi^2 cB = 3.43 \times 10^{-46}$ kg m^2; $\mu = 8.725$ amu.

ANSWER $B = 0.816$ **cm**$^{-1}$, $I = 3.43 \times 10^{-46}$ **kg m²**, $r_0 = 1.538$ **Å**

Q1.P1.6 1. $B = 21.18/2 = 10.59$ cm^{-1}; $I = h/8\pi^2 cB = 2.64 \times 10^{-47}$ kg m^2; $\mu = 0.9801$ amu.

ANSWER $B = 10.59$ **cm**$^{-1}$, $I = 2.64 \times 10^{-47}$ **kg m²**, $r_0 = 1.274$ **Å**

Set 2 · Moment of inertia and line spacing from B

Q1.P2.1 1. $I = h/8\pi^2 cB$; spacing $= 2B$; lines at $2B, 4B, 6B$.

ANSWER $I = 2.65 \times 10^{-47}$ kg m²; spacing 21.12 cm^{-1}; lines $21.12, 42.24, 63.36$ cm^{-1}

Q1.P2.2 1. $I = h/8\pi^2 cB$; spacing $= 2B$; lines at $2B, 4B, 6B$.

ANSWER $I = 5.59 \times 10^{-47}$ kg m²; spacing 10 cm^{-1}; lines $10, 20, 30$ cm^{-1}

Q1.P2.3 1. $I = h/8\pi^2 cB$; spacing $= 2B$; lines at $2B, 4B, 6B$.

ANSWER $I = 1.45 \times 10^{-46}$ kg m²; spacing 3.86 cm^{-1}; lines $3.86, 7.72, 11.58$ cm^{-1}

Q1.P2.4 1. $I = h/8\pi^2 cB$; spacing $= 2B$; lines at $2B, 4B, 6B$.

ANSWER $I = 3.29 \times 10^{-47}$ kg m²; spacing 17 cm^{-1}; lines $17, 34, 51$ cm^{-1}

Q1.P2.5 1. $I = h/8\pi^2 cB$; spacing $= 2B$; lines at $2B, 4B, 6B$.

ANSWER $I = 7.17 \times 10^{-46}$ kg m²; spacing 0.78 cm^{-1}; lines $0.78, 1.56, 2.34$ cm^{-1}

Set 3 · Position of a high-J line

Q1.P3.1 1. $\Delta\varepsilon = 2B(J+1) = (J+1) \times$ (first line); adjacent-line gap $= 2B$.

ANSWER 434.28 cm^{-1}; gap $= 20.68$ cm^{-1}

Q1.P3.2 1. $\Delta\varepsilon = 2B(J+1) = (J+1) \times$ (first line); adjacent-line gap $= 2B$.

ANSWER 211.80 cm^{-1}; gap $= 21.18$ cm^{-1}

Q1.P3.3 1. $\Delta\varepsilon = 2B(J+1) = (J+1) \times$ (first line); adjacent-line gap $= 2B$.

ANSWER 257.85 cm^{-1}; gap $= 17.19$ cm^{-1}

Q1.P3.4 1. $\Delta\varepsilon = 2B(J+1) = (J+1) \times$ (first line); adjacent-line gap $= 2B$.

ANSWER 65.11 cm^{-1}; gap $= 13.022$ cm^{-1}

Q1.P3.5 1. $\Delta\varepsilon = 2B(J+1) = (J+1) \times$ (first line); adjacent-line gap $= 2B$.

ANSWER 115.80 cm^{-1}; gap $= 3.86$ cm^{-1}

Set 4 · The reverse problem: from I or r_o to the first line

Q1.P4.1 1. If r_0 is given: $I = \mu r_0^2$. Then $B = h/8\pi^2 Ic$, first line $= 2B$.

ANSWER $B = 8.581$ cm^{-1}, first line $= 17.16$ cm^{-1}

Q1.P4.2 1. If r_0 is given: $I = \mu r_0^2$. Then $B = h/8\pi^2 Ic$, first line $= 2B$.

ANSWER $B = 10.596$ cm^{-1}, first line $= 21.19$ cm^{-1}

Q1.P4.3 1. If r_0 is given: $I = \mu r_0^2$. Then $B = h/8\pi^2 Ic$, first line $= 2B$.

ANSWER $B = 6.501$ **cm**$^{-1}$**, first line** $= 13.00$ **cm**$^{-1}$

Q1.P4.4 1. If r_0 is given: $I = \mu r_0^2$. Then $B = h/8\pi^2 Ic$, first line $= 2B$.

ANSWER $B = 1.932$ **cm**$^{-1}$**, first line** $= 3.86$ **cm**$^{-1}$

Q1.P4.5 1. If r_0 is given: $I = \mu r_0^2$. Then $B = h/8\pi^2 Ic$, first line $= 2B$.

ANSWER $B = 20.936$ **cm**$^{-1}$**, first line** $= 41.87$ **cm**$^{-1}$

Set 5 · Concept twists

Q1.P5.1 1. r_0 is unchanged (electronic structure identical) so $I = \mu r_0^2$ and $B \propto 1/\mu$.
2. $\mu_{HCl} = 0.9801$, $\mu_{DCl} = 1.9057$ amu, so $B_{DCl} = B_{HCl}\,\mu_{HCl}/\mu_{DCl}$.

ANSWER **First line ≈ 10.89 cm**$^{-1}$ **— lower, because heavier μ gives larger I and smaller B.**

Q1.P5.2 1. Spacing $= 2B = 21.2$ so $B = 10.6$ cm^{-1}. Lines $= 2B(J+1)$: 42.4 → $J+1 = 2$ ($J = 1 \to 2$); 63.6 → 3; 84.8 → 4.

ANSWER $B = 10.6$ **cm**$^{-1}$**; transitions** $1 \to 2$, $2 \to 3$, $3 \to 4$

Q1.P5.3 1. Gross rule: permanent dipole moment. Specific rule: $\Delta J = \pm 1$.

ANSWER **Active: HCl, CO, HF, H_2O. Inactive: H_2, CO_2 (symmetric, zero net dipole), N_2, Cl_2.**

Q1.P5.4 1. Level J has $(2J+1)$ states (degeneracy) but its population also falls exponentially as energy rises (Boltzmann factor).

ANSWER **Degeneracy** $(2J+1)$ **grows; Boltzmann population** $e^{-E_J/kT}$ **decays; the product peaks at an intermediate** J**.**

Q1.P5.5 1. Lines at $21.12\,(J+1)$; $21.12n < 100 \Rightarrow n \leq 4$.

ANSWER **4 lines (21.12, 42.24, 63.36, 84.48 cm**$^{-1}$**)**

CHAPTER 2 · TOPICS T5

Vibrational (IR) spectroscopy – harmonic oscillator

8 solved past questions | 19 practice questions | 4 theory answers

Every paper has a k ↔ $\bar{\nu}$ calculation, usually with zero-point energy, plus the 'why only one line' reason. Learn the one-line formula and practise the unit handling.

RECAP · FORMULAS YOU NEED

$$\bar{\nu}_{osc} = \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}}\ \text{cm}^{-1}$$

$$k = 4\pi^2 c^2 \bar{\nu}^2 \mu$$

$$\varepsilon_v = \left(v + \tfrac{1}{2}\right)\bar{\nu}_{osc}$$

$$\text{ZPE} = \varepsilon_0 = \tfrac{1}{2}\bar{\nu}_{osc}\ \ (E_0 = \tfrac{1}{2}hc\bar{\nu})$$

$$\Delta v = \pm 1 \ \Rightarrow\ \text{one line at } \bar{\nu}_{osc}$$

METHOD

1. Find μ in kg.
2. $\bar{\nu} = \dfrac{1}{2\pi c}\sqrt{k/\mu}$ (c in cm s^{-1} → cm^{-1}).
3. ZPE $= \frac{1}{2}\bar{\nu}$ (multiply by hc for joules).
4. For k: $k = 4\pi^2 c^2 \bar{\nu}^2 \mu$.

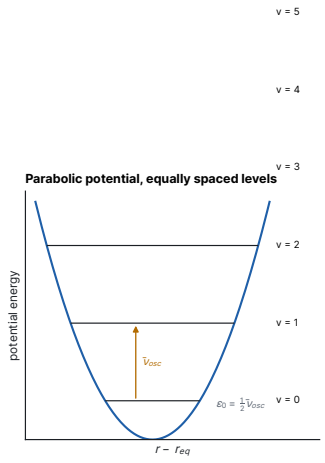


Fig. G13

A · PAST ESA QUESTIONS, FULLY SOLVED

Q2.1 (i) The force constant of HF (SHO) is 970 N m^{-1}. Calculate the oscillation frequency (cm^{-1}) and the zero-point energy. (ii) Why does the SHO spectrum show only one line? (c = 3×10^8 m s^{-1}, π = 3.14, 1 amu = 1.66×10^{-27} kg; H = 1.008, F = 18.99 amu)

Jun-24 · UE23 · Set 1 · 6 marks

1. $\mu = \dfrac{1.008 \times 18.99}{19.998} = 0.9572 \text{ amu} = 1.59 \times 10^{-27} \text{ kg}.$
2. $\bar{\nu} = \dfrac{1}{2\pi c}\sqrt{\dfrac{k}{\mu}} = \dfrac{1}{2(3.14)(3 \times 10^{10})}\sqrt{\dfrac{970}{1.59 \times 10^{-27}}} = 4147 \text{ cm}^{-1}.$
3. ZPE $= \frac{1}{2}\bar{\nu} = 2074 \text{ cm}^{-1} = 4.12e-20 \text{ J}$ ($\times hc$).
4. (ii) Levels $\varepsilon_v = (v + \frac{1}{2})\bar{\nu}$ are equally spaced and the selection rule is $\Delta v = \pm 1$, so every allowed transition needs the same energy $\bar{\nu}$ — one line.

ANSWER $\bar{\nu} \approx 4147$ **cm^{-1}, ZPE ≈ 2074 cm^{-1} ≈ 4.12e-20 J**

Q2.2 Same HF question with H = 1.004 amu, F = 18.998 amu (k = 970 N m^{-1}): (i) oscillation frequency in cm^{-1}; (ii) zero-point energy; (iii) why one line?

Dec-24 · UE22 · 6 marks

1. $\mu = \dfrac{1.004 \times 18.998}{20.002} = 0.9536 \text{ amu} = 1.58 \times 10^{-27} \text{ kg}.$
2. $\bar{\nu} = \dfrac{1}{2\pi c}\sqrt{\dfrac{k}{\mu}} = \dfrac{1}{2(3.14)(3 \times 10^{10})}\sqrt{\dfrac{970}{1.58 \times 10^{-27}}} = 4155 \text{ cm}^{-1}.$
3. ZPE $= \frac{1}{2}\bar{\nu} = 2077 \text{ cm}^{-1} = 4.13e-20 \text{ J}$ ($\times hc$).
4. (iii) Equal spacing + $\Delta v = \pm 1$.

ANSWER $\bar{\nu} \approx 4155$ **cm^{-1}; ZPE ≈ 2077 cm^{-1}**

Q2.3 (i) CO absorbs at 2168 cm^{-1} in a vibrational transition. Calculate the ground-state vibrational energy and force constant (SHO). ($c = 3 \times 10^{10}$ cm s^{-1}, $h = 6.625 \times 10^{-34}$ J s, C = 12, O = 15.999 amu). (ii) Write a note on the failure of the SHO model.

Jun-24 · UE23 · Set 2 · 6 marks

1. Ground state ($v = 0$): $\varepsilon_0 = \frac{1}{2}(2168) = 1084 \text{ cm}^{-1} = 1084 \times hc = 2.15e-20 \text{ J}.$
2. $\mu = 6.8570$ amu; $k = 4\pi^2 c^2 \bar{\nu}^2 \mu = 1.90 \times 10^3 \approx 1899 \text{ N m}^{-1}.$
3. (ii) See Q3 and the note after this chapter: real bonds dissociate, levels converge, overtones/hot bands occur — SHO predicts none.

ANSWER $\varepsilon_0 = 1084$ **cm^{-1}**; $k \approx 1899$ **N m^{-1}**

Q2.4 (i) Radiation absorbed by CO in a vibrational transition is 2168 cm^{-1}. Calculate the ground-state vibrational energy and the force constant (SHO; π = 3.14, h = 6.627×10^{-34} J s, c = 3×10^{10} cm s^{-1}, C = 12, O = 15.999). (ii) Only one peak is observed in the SHO model. Substantiate.

Jan · UE23A · Set 2 · 6 marks

1. Same as the previous CO question: $\varepsilon_0 = 1084\ \text{cm}^{-1}$; $k \approx 1.90 \times 10^3\ \text{N m}^{-1}$.
2. (ii) Equal level spacing and $\Delta v = \pm 1$.

ANSWER **ZPE 1084 cm^{-1}; k ≈ 1.90×10^3 N m^{-1}**

Q2.5 If the IR spectrum of CO shows only one line at 2050 cm^{-1}, calculate the force constant of CO. (C = 12, O = 15.9994 amu; 1 amu $= 1.66 \times 10^{-27}$ kg, $c = 3 \times 10^8\ \text{m s}^{-1}$)

Jul-23 · UE22 · B · 4 marks

1. $\mu = 6.8570\ \text{amu} = 1.14 \times 10^{-26}\ \text{kg}$; $\bar{\nu} = 2050\ \text{cm}^{-1} = 205000\ \text{m}^{-1}$.
2. $k = 4\pi^2 c^2 \bar{\nu}^2 \mu = 1.70 \times 10^3\ \text{N m}^{-1}$.

ANSWER $k \approx 1700$ **N m^{-1}**

Q2.6 For HBr (SHO) the vibrational spectrum shows a fundamental at 2649 cm^{-1}. Determine the force constant. (Reduced mass of HBr $= 1.639 \times 10^{-27}$ kg, $c = 3 \times 10^8\ \text{m s}^{-1}$)

Jan–May-24 · UE22 · 6 marks

1. μ is given — no need to compute it.
2. $k = 4\pi^2 c^2 \bar{\nu}^2 \mu$ with $\bar{\nu} = 264900\ \text{m}^{-1}$: $k = 4.09 \times 10^2 \approx 409\ \text{N m}^{-1}$.

ANSWER $k \approx 409$ **N m^{-1}**

Q2.7 The force constant of CO is 1840 N m^{-1}. Calculate the oscillation frequency and wavenumber. (1 amu $= 1.66 \times 10^{-27}$ kg, $c = 3 \times 10^8\ \text{m s}^{-1}$; C = 12.000, O = 15.9994)

Dec-24 · UE23 · 4 marks

1. $\mu = \dfrac{12.0 \times 15.9994}{27.999} = 6.8570\ \text{amu} = 1.14 \times 10^{-26}\ \text{kg}$.
2. $\bar{\nu} = \dfrac{1}{2\pi c}\sqrt{\dfrac{k}{\mu}} = \dfrac{1}{2(3.14)(3 \times 10^{10})}\sqrt{\dfrac{1840}{1.14 \times 10^{-26}}} = 2133\ \text{cm}^{-1}$.
3. Frequency $\nu = c\bar{\nu} = (3 \times 10^{10})(2133) = 6.40 \times 10^{13}$ Hz.

ANSWER $\bar{\nu} \approx 2133$ **cm^{-1}**; $\nu \approx 6.40 \times 10^{13}$ **Hz**

Q2.8 (i) Using the harmonic oscillator model calculate the zero-point vibrational energy of NO (k = 1530 N m^{-1}; N = 14, O = 16 amu). (ii) Draw the vibrational level diagram and the spectrum. (iii) Calculate the energies of $v = 3$ and $v = 4$. ($c = 3 \times 10^8$ m s^{-1}, $\pi = 3.14$, $h = 6.627 \times 10^{-34}$ J s)

Jan-25 · UE24 · Set 1 · 7 marks

1. $\mu = \dfrac{14 \times 16}{30} = 7.4667$ amu $= 1.24 \times 10^{-26}$ kg.
2. $\bar{\nu} = \dfrac{1}{2\pi c}\sqrt{\dfrac{k}{\mu}} = \dfrac{1}{2(3.14)(3 \times 10^{10})}\sqrt{\dfrac{1530}{1.24 \times 10^{-26}}} = 1865$ cm^{-1}.
3. ZPE $= \frac{1}{2}\bar{\nu} = 932$ $cm^{-1} = 1.85e - 20$ J ($\times hc$).
4. $\bar{\nu} = 1865$ cm^{-1}; ZPE $= 932$ $cm^{-1} = 1.85e - 20$ J.
5. (ii) Equally spaced levels (gap $\bar{\nu}$) on a parabola; one line at $\bar{\nu}$ (**Fig. G13**).
6. (iii) $\varepsilon_3 = 3.5\bar{\nu} = 6527$ cm^{-1}; $\varepsilon_4 = 4.5\bar{\nu} = 8392$ cm^{-1}.

ANSWER **ZPE ≈ 932 cm^{-1}; ε_3 ≈ 6527, ε_4 ≈ 8392 cm^{-1}**

THEORY QUESTIONS · MODEL ANSWERS

Why does the SHO spectrum show only one line? (asked ×5)

Levels $\varepsilon_v = (v + \frac{1}{2})\bar{\nu}$ are **equally spaced**; the selection rule $\Delta v = \pm 1$ allows only neighbouring-level jumps. Every jump has the same energy $\bar{\nu}$, so all coincide in one line.

Why is CO_2 IR-active though it has no permanent dipole? (asked ×3)

IR activity needs a **change** in dipole during vibration, not a permanent dipole. CO_2's symmetric stretch gives no change (inactive), but the asymmetric stretch and the bending mode do (active). It has no permanent dipole, so it is microwave-inactive.

Why is HI IR-active but H_2 and I_2 inactive?

HI is heteronuclear: stretching changes its dipole moment. H_2 and I_2 are homonuclear: zero dipole at every bond length, so the vibration never changes it.

Write a note on the failure of the SHO model. (asked ×1, relevant ×4)

The SHO assumes a parabolic potential and equal spacing, so it predicts a single line and never dissociation. Real bonds break when stretched (anharmonic, Morse), levels converge, $\Delta v = \pm 2, \pm 3$ overtones and hot bands appear, and the molecule can dissociate. The SHO therefore fails at high v and cannot explain overtones.

B · PRACTICE QUESTIONS (ANSWERS IN THE KEY AT THE END OF THIS CHAPTER)

Set 1 · k → wavenumber and zero-point energy

Exam pattern: HF, NO, CO k-values. Practise with other diatomics.

Q2.P1.1 The force constant of HCl is 516 N m^{-1}. Calculate the vibrational wavenumber (cm^{-1}), the frequency (Hz) and the zero-point energy in cm^{-1} and joules. ($c = 3 \times 10^{10}$ cm s^{-1}, $h = 6.626 \times 10^{-34}$ J s, 1 amu $= 1.66 \times 10^{-27}$ kg; masses 1.008, 35.45 amu)

Q2.P1.2 The force constant of HBr is 412 N m^{-1}. Calculate the vibrational wavenumber (cm^{-1}), the frequency (Hz) and the zero-point energy in cm^{-1} and joules. ($c = 3 \times 10^{10}$ cm s^{-1}, $h = 6.626 \times 10^{-34}$ J s, 1 amu $= 1.66 \times 10^{-27}$ kg; masses 1.008, 79.9 amu)

Q2.P1.3 The force constant of HI is 314 N m^{-1}. Calculate the vibrational wavenumber (cm^{-1}), the frequency (Hz) and the zero-point energy in cm^{-1} and joules. ($c = 3 \times 10^{10}$ cm s^{-1}, $h = 6.626 \times 10^{-34}$ J s, 1 amu $= 1.66 \times 10^{-27}$ kg; masses 1.008, 126.9 amu)

Q2.P1.4 The force constant of CO is 1857 N m^{-1}. Calculate the vibrational wavenumber (cm^{-1}), the frequency (Hz) and the zero-point energy in cm^{-1} and joules. ($c = 3 \times 10^{10}$ cm s^{-1}, $h = 6.626 \times 10^{-34}$ J s, 1 amu $= 1.66 \times 10^{-27}$ kg; masses 12.0, 15.995 amu)

Q2.P1.5 The force constant of HF is 966 N m^{-1}. Calculate the vibrational wavenumber (cm^{-1}), the frequency (Hz) and the zero-point energy in cm^{-1} and joules. ($c = 3 \times 10^{10}$ cm s^{-1}, $h = 6.626 \times 10^{-34}$ J s, 1 amu $= 1.66 \times 10^{-27}$ kg; masses 1.008, 18.998 amu)

Q2.P1.6 The force constant of NO is 1595 N m^{-1}. Calculate the vibrational wavenumber (cm^{-1}), the frequency (Hz) and the zero-point energy in cm^{-1} and joules. ($c = 3 \times 10^{10}$ cm s^{-1}, $h = 6.626 \times 10^{-34}$ J s, 1 amu $= 1.66 \times 10^{-27}$ kg; masses 14.0, 16.0 amu)

Set 2 · $\bar{\nu}$ → force constant

The reverse calculation. A stiffer bond (larger k) gives a higher $\bar{\nu}$.

Q2.P2.1 The fundamental IR band of HCl (treated as an SHO) is at 2886 cm^{-1}. Calculate the force constant. (1.008 and 35.45 amu; $c = 3 \times 10^{10}$ cm s^{-1})

Q2.P2.2 The fundamental IR band of CO (treated as an SHO) is at 2143 cm^{-1}. Calculate the force constant. (12.0 and 15.995 amu; $c = 3 \times 10^{10}$ cm s^{-1})

Q2.P2.3 The fundamental IR band of HF (treated as an SHO) is at 4138 cm^{-1}. Calculate the force constant. (1.008 and 18.998 amu; $c = 3 \times 10^{10}$ cm s^{-1})

Q2.P2.4 The fundamental IR band of NO (treated as an SHO) is at 1904 cm^{-1}. Calculate the force constant. (14.0 and 16.0 amu; $c = 3 \times 10^{10}$ cm s^{-1})

Q2.P2.5 The fundamental IR band of HI (treated as an SHO) is at 2309 cm^{-1}. Calculate the force constant. (1.008 and 126.9 amu; $c = 3 \times 10^{10}$ cm s^{-1})

Set 3 · Energy of the v-th level

Draw the ladder yourself first; the exam also asks for the level diagram and the single-line spectrum.

Q2.P3.1 For NO ($k = 1530$ N m^{-1}) calculate the energies of levels $v = 0, 1, 2, 3, 4$ in cm^{-1}, and the spacing between neighbouring levels.

Q2.P3.2 For HCl ($k = 516$ N m^{-1}) calculate the energies of levels $v = 0, 1, 2, 5$ in cm^{-1}, and the spacing between neighbouring levels.

Q2.P3.3 For CO ($k = 1857$ N m^{-1}) calculate the energies of levels $v = 0, 2, 3, 6$ in cm^{-1}, and the spacing between neighbouring levels.

Q2.P3.4 For HF ($k = 966$ N m^{-1}) calculate the energies of levels $v = 1, 2, 4$ in cm^{-1}, and the spacing between neighbouring levels.

Set 4 · Concept twists

Q2.P4.1 HCl and DCl have the same force constant (516 N m^{-1}). Compare their fundamental wavenumbers and ZPE.

Q2.P4.2 Which of the following are IR active: N_2, HCl, CO_2 (symmetric stretch), CO_2 (bend), H_2O (bend), O_2, CH_4 (all stretches)?

Q2.P4.3 If k is doubled, by what factor does the SHO wavenumber change? If μ is doubled?

Q2.P4.4 State the gross and specific selection rules for IR absorption of a diatomic.

C · ANSWER KEY

Set 1 · k → wavenumber and zero-point energy

Q2.P1.1 1. $\mu = 0.9801$ amu; $\bar{\nu} = (1/2\pi c)\sqrt{k/\mu}$; ZPE $= \frac{1}{2}\bar{\nu}$.

ANSWER $\bar{\nu} = 2988$ **cm**$^{-1}$; $\nu = 8.96 \times 10^{13}$ **Hz; ZPE** $= 1494$ **cm**$^{-1} = 2.97 \times 10^{-20}$ **J**

Q2.P1.2 1. $\mu = 0.9954$ amu; $\bar{\nu} = (1/2\pi c)\sqrt{k/\mu}$; ZPE $= \frac{1}{2}\bar{\nu}$.

ANSWER $\bar{\nu} = 2649$ **cm**$^{-1}$; $\nu = 7.95 \times 10^{13}$ **Hz; ZPE** $= 1325$ **cm**$^{-1} = 2.63 \times 10^{-20}$ **J**

Q2.P1.3 1. $\mu = 1.0001$ amu; $\bar{\nu} = (1/2\pi c)\sqrt{k/\mu}$; ZPE $= \frac{1}{2}\bar{\nu}$.

ANSWER $\bar{\nu} = 2307$ **cm**$^{-1}$; $\nu = 6.92 \times 10^{13}$ **Hz; ZPE** $= 1154$ **cm**$^{-1} = 2.29 \times 10^{-20}$ **J**

Q2.P1.4 1. $\mu = 6.8562$ amu; $\bar{\nu} = (1/2\pi c)\sqrt{k/\mu}$; ZPE $= \frac{1}{2}\bar{\nu}$.

ANSWER $\bar{\nu} = 2143$ **cm**$^{-1}$; $\nu = 6.43 \times 10^{13}$ **Hz; ZPE** $= 1071$ **cm**$^{-1} = 2.13 \times 10^{-20}$ **J**

Q2.P1.5 1. $\mu = 0.9572$ amu; $\bar{\nu} = (1/2\pi c)\sqrt{k/\mu}$; ZPE $= \frac{1}{2}\bar{\nu}$.

ANSWER $\bar{\nu} = 4136$ **cm**$^{-1}$**;** $\nu = 1.24 \times 10^{14}$ **Hz; ZPE** $= 2068$ **cm**$^{-1} = 4.11 \times 10^{-20}$ **J**

Q2.P1.6 1. $\mu = 7.4667$ amu; $\bar{\nu} = (1/2\pi c)\sqrt{k/\mu}$; ZPE $= \frac{1}{2}\bar{\nu}$.

ANSWER $\bar{\nu} = 1903$ **cm**$^{-1}$**;** $\nu = 5.71 \times 10^{13}$ **Hz; ZPE** $= 952$ **cm**$^{-1} = 1.89 \times 10^{-20}$ **J**

Set 2 · ν̄ → force constant

Q2.P2.1 1. $k = 4\pi^2 c^2 \bar{\nu}^2 \mu$, with μ in kg.

ANSWER $k = 4.81 \times 10^2 \approx 481$ **N m**$^{-1}$

Q2.P2.2 1. $k = 4\pi^2 c^2 \bar{\nu}^2 \mu$, with μ in kg.

ANSWER $k = 1.86 \times 10^3 \approx 1857$ **N m**$^{-1}$

Q2.P2.3 1. $k = 4\pi^2 c^2 \bar{\nu}^2 \mu$, with μ in kg.

ANSWER $k = 9.67 \times 10^2 \approx 967$ **N m**$^{-1}$

Q2.P2.4 1. $k = 4\pi^2 c^2 \bar{\nu}^2 \mu$, with μ in kg.

ANSWER $k = 1.60 \times 10^3 \approx 1597$ **N m**$^{-1}$

Q2.P2.5 1. $k = 4\pi^2 c^2 \bar{\nu}^2 \mu$, with μ in kg.

ANSWER $k = 3.14 \times 10^2 \approx 314$ **N m**$^{-1}$

Set 3 · Energy of the v-th level

Q2.P3.1 1. $\varepsilon_v = (v + \frac{1}{2})\bar{\nu}$; spacing $= \bar{\nu}$.

ANSWER $\bar{\nu} = 1864$**:** $\varepsilon_0 = 932$**,** $\varepsilon_1 = 2796$**,** $\varepsilon_2 = 4660$**,** $\varepsilon_3 = 6524$**,** $\varepsilon_4 = 8388$ **cm**$^{-1}$**; spacing 1864 cm**$^{-1}$

Q2.P3.2 1. $\varepsilon_v = (v + \frac{1}{2})\bar{\nu}$; spacing $= \bar{\nu}$.

ANSWER $\bar{\nu} = 2988$**:** $\varepsilon_0 = 1494$**,** $\varepsilon_1 = 4481$**,** $\varepsilon_2 = 7469$**,** $\varepsilon_5 = 16432$ **cm**$^{-1}$**; spacing 2988 cm**$^{-1}$

Q2.P3.3 1. $\varepsilon_v = (v + \frac{1}{2})\bar{\nu}$; spacing $= \bar{\nu}$.

ANSWER $\bar{\nu} = 2143$**:** $\varepsilon_0 = 1071$**,** $\varepsilon_2 = 5357$**,** $\varepsilon_3 = 7500$**,** $\varepsilon_6 = 13929$ **cm**$^{-1}$**; spacing 2143 cm**$^{-1}$

Q2.P3.4 1. $\varepsilon_v = (v + \frac{1}{2})\bar{\nu}$; spacing $= \bar{\nu}$.

ANSWER $\bar{\nu} = 4136$**:** $\varepsilon_1 = 6205$**,** $\varepsilon_2 = 10341$**,** $\varepsilon_4 = 18614$ **cm**$^{-1}$**; spacing 4136 cm**$^{-1}$

Set 4 · Concept twists

Q2.P4.1 1. $\bar{\nu} \propto 1/\sqrt{\mu}$: $\mu_{HCl} = 0.9801$, $\mu_{DCl} = 1.9057$ amu.

ANSWER HCl 2988 cm⁻¹ (ZPE 1494); DCl 2143 cm⁻¹ (ZPE 1071). Ratio 1.394. Heavier isotope → lower ν̄ and ZPE.

Q2.P4.2 1. Test: does the dipole moment change during the vibration?

ANSWER Active: HCl, CO_2 (bend), H_2O (bend), CH_4 (some stretches). Inactive: N_2, O_2, CO_2 symmetric stretch, CH_4 totally symmetric stretch.

Q2.P4.3 1. $\bar{\nu} \propto \sqrt{k/\mu}$.

ANSWER ×√2 = 1.414 when k doubles; ×1/√2 = 0.707 when μ doubles.

Q2.P4.4 1. Gross: dipole moment must change on vibration. Specific (SHO): $\Delta v = \pm 1$.

ANSWER Gross: change in dipole; Specific: Δv = ±1 (SHO), Δv = ±1, ±2, ±3… (anharmonic).

CHAPTER 3 · TOPICS T6

Anharmonic oscillator and overtones

5 solved past questions | 18 practice questions | 0 theory answers

Anharmonic problems are two-equation puzzles: you are given two of fundamental / overtone / ω_e / x_e and must find the others. Memorise the three transition expressions.

RECAP · FORMULAS YOU NEED

$$\varepsilon_v = (v+\tfrac{1}{2})\bar{\omega}_e - (v+\tfrac{1}{2})^2\bar{\omega}_e x_e$$

$$0 \to 1 :\ \bar{\omega}_e(1-2x_e)$$

$$0 \to 2 :\ 2\bar{\omega}_e(1-3x_e)$$

$$0 \to 3 :\ 3\bar{\omega}_e(1-4x_e)$$

$$1 \to 2\ (\text{hot}) :\ \bar{\omega}_e(1-4x_e)$$

METHOD

1. Write the two given transitions as two equations in $\bar{\omega}_e$ and x_e.
2. Subtract: $\bar{\omega}_e x_e = f - o/2$; then $\bar{\omega}_e = 3f - o$ and $x_e = (f - o/2)/\bar{\omega}_e$.
3. If x_e is given instead: $\bar{\omega}_e = f/(1-2x_e)$.

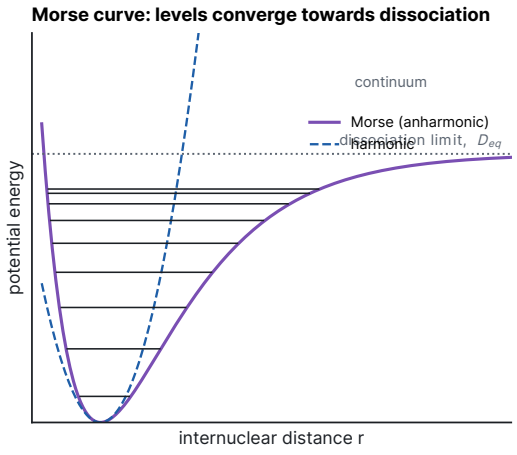


Fig. G15

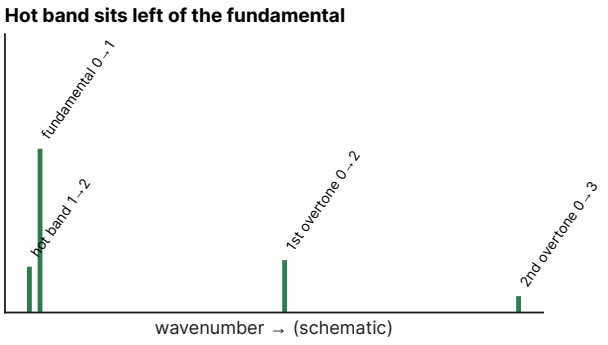


Fig. G17

A · PAST ESA QUESTIONS, FULLY SOLVED

Q3.1 For HBr the vibrational spectrum shows the fundamental at 2559.08 cm^{-1} and the first overtone at 5027.54 cm^{-1}. Calculate the anharmonicity constant (anharmonic model).

Jul-23 · UE22 · A | 4 marks

1. Fundamental: $\bar{\omega}_e(1-2x_e) = 2559.08$. First overtone: $2\bar{\omega}_e(1-3x_e) = 5027.54 \Rightarrow \bar{\omega}_e(1-3x_e) = 2513.77$.
2. Subtract: $\bar{\omega}_e x_e = 2559.08 - 2513.77 = 45.31\ \text{cm}^{-1}$. Then $\bar{\omega}_e = 2559.08 + 2(45.31) = 3(2559.08) - 5027.54 = 2649.70\ \text{cm}^{-1}$.
3. $x_e = 45.31/2649.70 = 0.0171$.

ANSWER $x_e \approx 0.0171$ ($\bar{\omega}_e \approx 2649.7\ \text{cm}^{-1}$)

Q3.2 The fundamental stretching frequency of CO is 2143 cm^{-1} and its anharmonicity constant is 0.01. Calculate its oscillational (equilibrium) frequency.

Jan-25 · UE24 · Set 2 | 2 marks

1. $\bar{\omega}_e(1-2x_e) = 2143$.
2. $\bar{\omega}_e = \dfrac{2143}{1-2(0.01)} = \dfrac{2143}{0.98} = 2186.7\ \text{cm}^{-1}$.

ANSWER $\bar{\omega}_e \approx 2186.7\ \text{cm}^{-1}$

Q3.3 Assuming CO shows anharmonic oscillation, calculate the wavenumber of the first overtone if $x_e = 0.017$ and the equilibrium vibrational frequency is 2050 cm^{-1}.

Jul-23 · UE22 · B | 2 marks

1. First overtone $0 \to 2$: $2\bar{\omega}_e(1-3x_e) = 2(2050)(1-3\times 0.017) = 2(2050)(0.949)$.

ANSWER $3890.9\ \text{cm}^{-1}$

Q3.4 (i) What is the selection rule for the anharmonic oscillator? What is the first overtone transition? Give its expression and explain the terms.

Jan-25 · UE24 · Set 2 | 4 marks

1. Selection rule: $\Delta v = \pm 1, \pm 2, \pm 3 \ldots$ (gross rule: dipole must change).
2. First overtone $v = 0 \to 2$: $\Delta\bar{\nu} = 2\bar{\omega}_e(1-3x_e)$, where $\bar{\omega}_e$ is the equilibrium vibrational wavenumber (cm^{-1}) and x_e is the small positive anharmonicity constant. It is a little less than twice the fundamental.

ANSWER $2\bar{\omega}_e(1-3x_e)$

Q3.5 Briefly describe the Born–Oppenheimer approximation and draw the ground and excited electronic states of a diatomic molecule showing rotational and vibrational energy levels (anharmonic model).

Dec-24 · UE23 · 5 marks

1. B–O: electronic, vibrational and rotational energies are independent; $E = E_{rot} + E_{vib} + E_{elec}$ (translation negligible), with $E_{rot} \ll E_{vib} \ll E_{elec}$.
2. Diagram: two Morse curves (ground below, excited above and displaced to longer r_e); levels converge towards each dissociation limit; each vibrational level carries closely spaced rotational sub-levels (**Fig. G15** for the Morse shape).

ANSWER **See statement + diagram**

B · PRACTICE QUESTIONS (ANSWERS IN THE KEY AT THE END OF THIS CHAPTER)

Set 1 · x_e and ω_e from fundamental + first overtone

The standard exam puzzle.

Q3.P1.1 HCl: fundamental 2885.3 cm^{-1}, first overtone 5665.0 cm^{-1}. Calculate x_e and $\bar{\omega}_e$.

Q3.P1.2 CO: fundamental 2143.2 cm^{-1}, first overtone 4259.9 cm^{-1}. Calculate x_e and $\bar{\omega}_e$.

Q3.P1.3 HF: fundamental 3958.5 cm^{-1}, first overtone 7737.2 cm^{-1}. Calculate x_e and $\bar{\omega}_e$.

Q3.P1.4 NO: fundamental 1876.1 cm^{-1}, first overtone 3724.0 cm^{-1}. Calculate x_e and $\bar{\omega}_e$.

Q3.P1.5 HBr: fundamental 2559.08 cm^{-1}, first overtone 5027.54 cm^{-1}. Calculate x_e and $\bar{\omega}_e$.

Set 2 · ω_e from the fundamental and x_e; overtones

Exam: CO 2143, x_e = 0.01. Same method, other molecules.

Q3.P2.1 CO: fundamental 2143 cm^{-1} and $x_e = 0.01$. Find $\bar{\omega}_e$ and the first and second overtones.

Q3.P2.2 HCl: fundamental 2886 cm^{-1} and $x_e = 0.0177$. Find $\bar{\omega}_e$ and the first and second overtones.

Q3.P2.3 HF: fundamental 3959 cm^{-1} and $x_e = 0.0217$. Find $\bar{\omega}_e$ and the first and second overtones.

Q3.P2.4 NO: fundamental 1876 cm^{-1} and $x_e = 0.0074$. Find $\bar{\omega}_e$ and the first and second overtones.

Q3.P2.5 HI: fundamental 2230 cm^{-1} and $x_e = 0.0183$. Find $\bar{\omega}_e$ and the first and second overtones.

Set 3 · All transitions from ω_e and x_e

Exam: CO ω_e = 2050, x_e = 0.017, first overtone.

Q3.P3.1 CO: $\bar{\omega}_e = 2050\ \text{cm}^{-1}$, $x_e = 0.017$. Calculate the fundamental, first overtone, second overtone and the hot band ($1 \to 2$).

Q3.P3.2 HCl: $\bar{\omega}_e = 2990\ \text{cm}^{-1}$, $x_e = 0.0177$. Calculate the fundamental, first overtone, second overtone and the hot band ($1 \to 2$).

Q3.P3.3 NO: $\bar{\omega}_e = 1904\ \text{cm}^{-1}$, $x_e = 0.0074$. Calculate the fundamental, first overtone, second overtone and the hot band ($1 \to 2$).

Q3.P3.4 HF: $\bar{\omega}_e = 4138\ \text{cm}^{-1}$, $x_e = 0.0217$. Calculate the fundamental, first overtone, second overtone and the hot band ($1 \to 2$).

Set 4 · Concept twists

Q3.P4.1 Why do the lines of an anharmonic spectrum get closer together at higher energy?

Q3.P4.2 Give the transition energy of the hot band relative to the fundamental for $\bar{\omega}_e = 2000$, $x_e = 0.02$.

Q3.P4.3 Compare SHO and anharmonic oscillator on: potential, level spacing, selection rule, number of lines.

Q3.P4.4 Calculate the anharmonic zero-point energy of CO ($\bar{\omega}_e = 2169.8$, $x_e = 0.00612$) and compare with $\frac{1}{2}\bar{\omega}_e$.

C · ANSWER KEY

Set 1 · x_e and ω_e from fundamental + first overtone

Q3.P1.1 1. $\bar{\omega}_e x_e = f - o/2 = 52.80$; $\bar{\omega}_e = 3f - o = 2990.9$.

ANSWER $\bar{\omega}_e = 2990.9\ \text{cm}^{-1}$, $x_e = 0.0177$

Q3.P1.2 1. $\bar{\omega}_e x_e = f - o/2 = 13.25$; $\bar{\omega}_e = 3f - o = 2169.7$.

ANSWER $\bar{\omega}_e = 2169.7\ \text{cm}^{-1}$, $x_e = 0.0061$

Q3.P1.3 1. $\bar{\omega}_e x_e = f - o/2 = 89.90$; $\bar{\omega}_e = 3f - o = 4138.3$.

ANSWER $\bar{\omega}_e = 4138.3\ \text{cm}^{-1}$, $x_e = 0.0217$

Q3.P1.4 1. $\bar{\omega}_e x_e = f - o/2 = 14.10$; $\bar{\omega}_e = 3f - o = 1904.3$.

ANSWER $\bar{\omega}_e = 1904.3\ \text{cm}^{-1}$, $x_e = 0.0074$

Q3.P1.5 1. $\bar{\omega}_e x_e = f - o/2 = 45.31$; $\bar{\omega}_e = 3f - o = 2649.7$.

ANSWER $\bar{\omega}_e = 2649.7$ cm^{-1}, $x_e = 0.0171$

Set 2 · ω_e from the fundamental and x_e; overtones

Q3.P2.1 1. $\bar{\omega}_e = f/(1 - 2x_e)$; overtones $2\bar{\omega}_e(1 - 3x_e)$ and $3\bar{\omega}_e(1 - 4x_e)$.

ANSWER $\bar{\omega}_e = 2186.7$; 1st overtone 4242.3; 2nd overtone 6297.8 cm^{-1}

Q3.P2.2 1. $\bar{\omega}_e = f/(1 - 2x_e)$; overtones $2\bar{\omega}_e(1 - 3x_e)$ and $3\bar{\omega}_e(1 - 4x_e)$.

ANSWER $\bar{\omega}_e = 2991.9$; 1st overtone 5666.1; 2nd overtone 8340.3 cm^{-1}

Q3.P2.3 1. $\bar{\omega}_e = f/(1 - 2x_e)$; overtones $2\bar{\omega}_e(1 - 3x_e)$ and $3\bar{\omega}_e(1 - 4x_e)$.

ANSWER $\bar{\omega}_e = 4138.6$; 1st overtone 7738.4; 2nd overtone 11338.2 cm^{-1}

Q3.P2.4 1. $\bar{\omega}_e = f/(1 - 2x_e)$; overtones $2\bar{\omega}_e(1 - 3x_e)$ and $3\bar{\omega}_e(1 - 4x_e)$.

ANSWER $\bar{\omega}_e = 1904.2$; 1st overtone 3723.8; 2nd overtone 5543.5 cm^{-1}

Q3.P2.5 1. $\bar{\omega}_e = f/(1 - 2x_e)$; overtones $2\bar{\omega}_e(1 - 3x_e)$ and $3\bar{\omega}_e(1 - 4x_e)$.

ANSWER $\bar{\omega}_e = 2314.7$; 1st overtone 4375.3; 2nd overtone 6435.8 cm^{-1}

Set 3 · All transitions from ω_e and x_e

Q3.P3.1 1. Use the four expressions in the recap.

ANSWER Fundamental 1980.3; 1st overtone 3890.9; 2nd overtone 5731.8; hot band 1910.6 cm^{-1} (hot band lies below the fundamental)

Q3.P3.2 1. Use the four expressions in the recap.

ANSWER Fundamental 2884.2; 1st overtone 5662.5; 2nd overtone 8334.9; hot band 2778.3 cm^{-1} (hot band lies below the fundamental)

Q3.P3.3 1. Use the four expressions in the recap.

ANSWER Fundamental 1875.8; 1st overtone 3723.5; 2nd overtone 5542.9; hot band 1847.6 cm^{-1} (hot band lies below the fundamental)

Q3.P3.4 1. Use the four expressions in the recap.

ANSWER Fundamental 3958.4; 1st overtone 7737.2; 2nd overtone 11336.5; hot band 3778.8 cm^{-1} (hot band lies below the fundamental)

Set 4 · Concept twists

Q3.P4.1 1. Levels converge towards the dissociation limit: spacing $= \bar{\omega}_e[1 - 2x_e(v + 1)]$ shrinks with v.

ANSWER Because the Morse levels converge; the gap between v and $v + 1$ decreases as v increases.

Q3.P4.2

1. Fundamental $2000(0.96) = 1920$; hot band $2000(0.92) = 1840$.

ANSWER **Hot band 1840 cm⁻¹ — 80 cm⁻¹ below the fundamental (the left side of the spectrum).**

Q3.P4.3

ANSWER **SHO: parabola, equal, Δv = ±1, one line. Anharmonic: Morse curve, converging, Δv = ±1, ±2, ±3…, fundamental + overtones (+ hot bands).**

Q3.P4.4

1. $\varepsilon_0 = \frac{1}{2}\bar{\omega}_e(1 - \frac{1}{2}x_e)$ (added for clarity).

ANSWER $\varepsilon_0 = 1081.6$ **cm**$^{-1}$ **vs 1084.9 — slightly lower.**

CHAPTER 4 · TOPICS T1–T3, T7

Born–Oppenheimer, electronic spectra, Franck–Condon and Raman

6 solved past questions | 14 practice questions | 7 theory answers

Largely theory: statements, diagrams and 'what can you conclude?' questions. Franck–Condon diagrams appear in more than half of all papers.

RECAP · FORMULAS YOU NEED

$$E = E_{rot} + E_{vib} + E_{elec}, \quad E_{rot} \ll E_{vib} \ll E_{elec}$$

$$E = h\nu = hc\bar{\nu}, \quad \bar{\nu} = 1/\lambda$$

electronic lines labelled (v', v'')

$$A = \log \frac{I_0}{I} = \varepsilon c l \quad \text{(Beer–Lambert; older syllabus)}$$

METHOD

1. Franck–Condon: always draw the vertical arrow from the centre of $v'' = 0$.
2. Compare r'_e with r''_e: equal → (0,0) strongest; larger → (v',0) with v'>0 strongest; much larger → continuum.

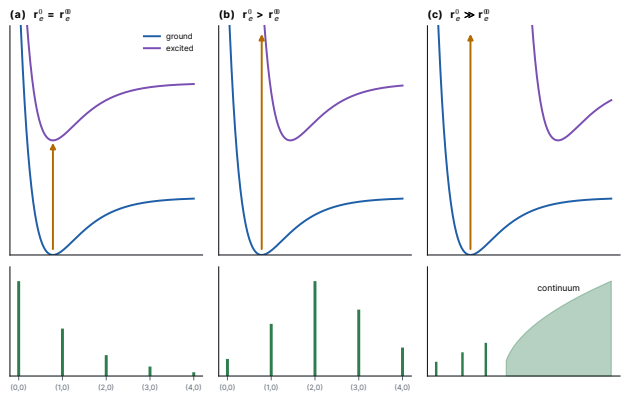


Fig. G21

A · PAST ESA QUESTIONS, FULLY SOLVED

Q4.1 State the Franck–Condon principle. With a diagram show the electronic transitions and spectra when (i) $v'' = 0 \rightarrow v' = 0$ is strongest; (ii) $v'' = 0 \rightarrow v' = 2$ is strongest.

Dec-24 · UE22 | 7 marks

1. Statement as in the theory box. (i) is case (a) of **Fig. G21** ($r'_e = r''_e$): (0,0) strongest. (ii) is case (b) ($r'_e > r''_e$): the vertical arrow ends on $v' = 2$, so the (2,0) line is strongest and the progression peaks there.

ANSWER Cases (a) and (b)

Q4.2 Justify the Franck–Condon principle by drawing the transitions and spectra when (1) the internuclear distances in ground and excited states are the same; (2) the excited-state distance is much larger.

Jun-24 · UE23 · Set 2 · 4 marks

1. (1) case (a): (0,0) most intense. (2) case (c): vertical arrow above the dissociation limit → continuum (**Fig. G21**).

ANSWER **Cases (a) and (c)**

Q4.3 (i) What is the Franck–Condon principle? If a vibrational-electronic spectrum shows a continuum, what can one infer? (ii) Why is bulk aluminium stable whereas nano aluminium is combustible?

Jul-23 · UE22 · B · 6 marks

1. (i) Statement + dissociation (theory box). (ii) is a nano-materials question (Unit 4) — not covered here.

ANSWER **Dissociation**

Q4.4 (i) State the Born–Oppenheimer approximation. (ii) What does the appearance of a continuum in an electronic spectrum let us conclude?

Jan–May-24 · UE22 · 6 marks

1. Statement and dissociation reasoning — see theory box.

ANSWER **See theory box**

Q4.5 What do you mean by Rayleigh scattering and Raman scattering? Mention the difference between Stokes and anti-Stokes lines.

Dec-23 · UE22 · 4 marks

1. See theory box.

ANSWER **See theory box**

Q4.6 How is it that carbon dioxide is IR active even though it has no permanent dipole? (also Jan-24 UE22 'Why is CO_2 IR active but microwave inactive?')

Jul-23 · UE22 · B · 2 marks

1. See Q2 theory box: asymmetric stretch and bend change the dipole; symmetric stretch does not.

ANSWER **Dipole *change*, not permanent dipole**

THEORY QUESTIONS · MODEL ANSWERS

State the Franck–Condon principle. Draw the three cases and comment on each spectrum. (asked ×5)

Statement: "An electronic transition takes place so rapidly that a vibrating molecule does not change its internuclear distance appreciably during the transition," so transitions are vertical.
(a) $r'_e = r''_e$: vertical arrow lands on $v' = 0$; the (0,0) line is strongest and intensity falls for (1,0), (2,0)...
(b) $r'_e > r''_e$: arrow lands on a higher v' (e.g. 2); the (v',0) line with v'>0 is strongest, so the maximum is displaced along the progression.
(c) $r'_e \gg r''_e$: arrow lands above the dissociation limit; a few discrete lines then a **continuum** (**Fig. G21**).

The vibrational–electronic spectrum of a molecule shows a continuum. What can be inferred? (asked ×2)

The molecule has been excited above the dissociation limit of the upper state, so it **dissociates** into atoms. The atoms carry away the excess energy as unquantised kinetic energy, so no discrete lines appear — a continuum. (It also implies $r'_e \gg r''_e$.)

State the Born–Oppenheimer approximation. (asked ×3)

The total energy of a molecule is the sum of independent translational, rotational, vibrational and electronic parts; translation is negligible, so $E = E_{rot} + E_{vib} + E_{elec}$ with $E_{rot} \ll E_{vib} \ll E_{elec}$ (electronic levels are far apart, rotational levels very close).

What are Rayleigh scattering and Raman scattering? Differentiate Stokes and anti-Stokes lines. (asked ×2)

Rayleigh: elastic scattering — scattered light has the same frequency as the incident light. **Raman:** inelastic scattering — frequency shifts because the molecule exchanges vibrational/rotational energy with the photon. **Stokes:** scattered frequency lower than incident (molecule gains energy, goes up). **Anti-Stokes:** higher than incident (molecule loses energy, comes down); weaker because fewer molecules start in an excited level. (Transcript-based; not on the 2026 slides.)

State the Beer–Lambert law and explain the terms. (older syllabus)

$A = \log(I_0/I) = \varepsilon c l$: absorbance is proportional to concentration c (mol L^{-1}) and path length l (cm); ε is the molar absorptivity (L mol^{-1} cm^{-1}); I_0 and I are incident and transmitted intensities. **Flag: not on current slides.**

Why does the excited electronic state usually have a larger equilibrium bond length?

An electron is promoted to an antibonding orbital (e.g. π → π*), weakening the bond, so the minimum of the upper potential curve shifts to larger r.

Why do homonuclear molecules (H_2, N_2) show vibration/rotation only in electronic spectra?

They have no dipole change for IR or rotation, but all molecules can undergo electronic transitions, which carry vibrational (coarse) and rotational (fine) structure.

B · PRACTICE QUESTIONS (ANSWERS IN THE KEY AT THE END OF THIS CHAPTER)

Set 1 · Wavelength, wavenumber, frequency, energy

Basic conversions that underpin every spectroscopy question.

Q4.P1.1 Light of wavelength 500 nm: find the wavenumber (cm^{-1}), frequency (Hz), photon energy (J) and region. ($h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1})

Q4.P1.2 Light of wavelength 250 nm: find the wavenumber (cm^{-1}), frequency (Hz), photon energy (J) and region. ($h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1})

Q4.P1.3 Light of wavelength 700 nm: find the wavenumber (cm^{-1}), frequency (Hz), photon energy (J) and region. ($h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1})

Q4.P1.4 Light of wavelength 10000 nm: find the wavenumber (cm^{-1}), frequency (Hz), photon energy (J) and region. ($h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1})

Q4.P1.5 Light of wavelength 3000 nm: find the wavenumber (cm^{-1}), frequency (Hz), photon energy (J) and region. ($h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^{10}$ cm s^{-1})

Set 2 · Franck–Condon outcomes

Decide the case, then describe the spectrum.

Q4.P2.1 A diatomic's ground and excited states have almost identical bond lengths. Which line is the most intense?

Q4.P2.2 The spectrum shows the strongest line as (3,0). Compare r_e' and r_e'' and sketch the transition.

Q4.P2.3 A spectrum has a few discrete lines that merge into a featureless band at high energy. What happened to the molecule?

Q4.P2.4 Why can't a vibrational spectrum of an electronic transition be explained without the Franck–Condon principle even though there is no vibrational selection rule?

Q4.P2.5 State two differences between the case where $r_e' = r_e''$ and $r_e' > r_e''$.

Set 3 · Selecting the right region and transition

Q4.P3.1 Which transition type and region: (a) spin reversal of nuclei; (b) rotation; (c) vibration; (d) valence-electron redistribution; (e) inner-electron; (f) nuclear rearrangement?

Q4.P3.2 Order rotation, vibration and electronic transitions by increasing energy and say which spectroscopy each belongs to.

Q4.P3.3 Name the three electronic transitions common in organic molecules and the orbitals involved.

Q4.P3.4 Draw/describe Stokes and anti-Stokes lines relative to the Rayleigh line.

C · ANSWER KEY

Set 1 · Wavelength, wavenumber, frequency, energy

Q4.P1.1 1. $\bar{\nu} = 1/\lambda$ (λ in cm); $\nu = c\bar{\nu}$; $E = h\nu$.

ANSWER $\bar{\nu} = 2.00 \times 10^{4}$ cm^{-1}; $\nu = 6.00 \times 10^{14}$ Hz; $E = 3.98 \times 10^{-19}$ J; region: visible

Q4.P1.2 1. $\bar{\nu} = 1/\lambda$ (λ in cm); $\nu = c\bar{\nu}$; $E = h\nu$.

ANSWER $\bar{\nu} = 4.00 \times 10^{4}$ cm^{-1}; $\nu = 1.20 \times 10^{15}$ Hz; $E = 7.95 \times 10^{-19}$ J; region: UV

Q4.P1.3 1. $\bar{\nu} = 1/\lambda$ (λ in cm); $\nu = c\bar{\nu}$; $E = h\nu$.

ANSWER $\bar{\nu} = 1.43 \times 10^{4}$ cm^{-1}; $\nu = 4.29 \times 10^{14}$ Hz; $E = 2.84 \times 10^{-19}$ J; region: visible

Q4.P1.4 1. $\bar{\nu} = 1/\lambda$ (λ in cm); $\nu = c\bar{\nu}$; $E = h\nu$.

ANSWER $\bar{\nu} = 1.00 \times 10^{3}$ cm^{-1}; $\nu = 3.00 \times 10^{13}$ Hz; $E = 1.99 \times 10^{-20}$ J; region: IR

Q4.P1.5 1. $\bar{\nu} = 1/\lambda$ (λ in cm); $\nu = c\bar{\nu}$; $E = h\nu$.

ANSWER $\bar{\nu} = 3.33 \times 10^{3}$ cm^{-1}; $\nu = 1.00 \times 10^{14}$ Hz; $E = 6.63 \times 10^{-20}$ J; region: IR

Set 2 · Franck–Condon outcomes

Q4.P2.1 **ANSWER** (0,0): case (a); intensity decreases for (1,0), (2,0)…

Q4.P2.2 **ANSWER** $r'_e > r''_e$ (case b): the vertical arrow from $v'' = 0$ ends on $v' = 3$.

Q4.P2.3 **ANSWER** It dissociated (case c); the continuum is unquantised kinetic energy of the atoms.

Q4.P2.4 **ANSWER** With no selection rule all (v′,v″) lines are allowed, yet they differ greatly in intensity. Franck–Condon (vertical transitions + nuclear probability) explains which are strong.

Q4.P2.5 **ANSWER** (a) one dominant (0,0) line, intensity falls quickly; (b) intensity maximum shifts to (v′,0) with v′>0 and the progression is longer.

Set 3 · Selecting the right region and transition

Q4.P3.1 **ANSWER** (a) radio/NMR 10 m–1 cm; (b) microwave 1 cm–100 μm; (c) IR 100 μm–1 μm; (d) UV-Vis 1 μm–10 nm; (e) X-ray 10 nm–100 pm; (f) γ-ray 100–1 pm.

Q4.P3.2 **ANSWER** $E_{rot} < E_{vib} < E_{elec}$: microwave < IR < UV-Vis.

Q4.P3.3 **ANSWER** σ→σ*, π→π*, n→π* (HOMO → LUMO).

Q4.P3.4 **ANSWER** Rayleigh at the incident frequency; Stokes at lower frequency (red side); anti-Stokes at higher frequency (blue side), weaker.

CHAPTER 5 · TOPICS T8–T12

Phase rule, water and Pb–Ag systems

9 solved past questions | 35 practice questions | 0 theory answers

Phase equilibria questions come in three flavours: derive the rule, count C/P/F for a system, and draw-and-label a diagram. Practise counting until it is automatic.

RECAP · FORMULAS YOU NEED

$$F = C - P + 2$$

$$\text{variables} = P(C-1) + 2, \quad \text{constraints} = C(P-1$$

$$\text{reduced: } F = C - P + 1$$

$$\text{water: } F = 3 - P$$

$$\frac{dP}{dT} = \frac{\Delta H}{T \Delta V}$$

METHOD

1. Count phases P (each solid, each immiscible liquid, all gases = 1).
2. Count components C = species − independent reactions − composition restrictions.
3. $F = C - P + 2$ (or $C - P + 1$ if pressure is fixed / condensed).
4. $F = 2$ bivariant, 1 univariant, 0 invariant.

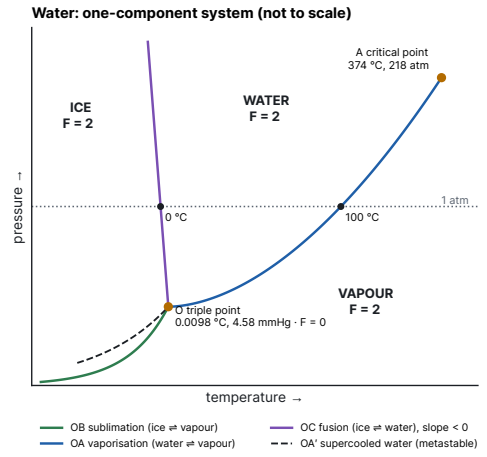


Fig. G22

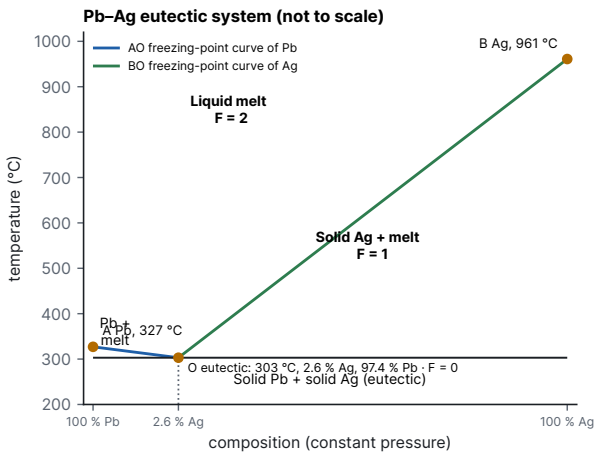


Fig. G23

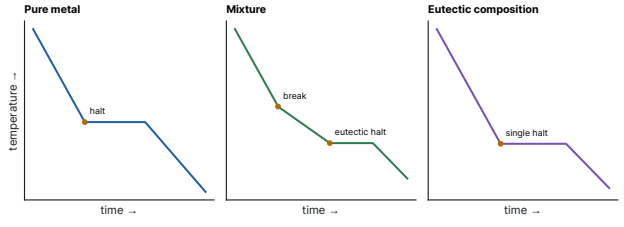


Fig. G24

Q5-D · DERIVATION OF THE PHASE RULE (ASKED ×3)

1. System of C components in P phases at equilibrium. Variables: temperature, pressure and the composition of each phase.
2. Composition of one phase needs $C-1$ mole fractions (they sum to 1). For P phases: $P(C-1)$. Add T and P: **total variables $= P(C-1)+2$**.
3. At equilibrium the chemical potential of each component is equal in every phase: $\mu_i^{\alpha} = \mu_i^{\beta} = \mu_i^{\gamma} = \cdots$ — that is $(P-1)$ equations per component, **$C(P-1)$ constraints** in all.
4. $F =$ variables − constraints $= P(C-1)+2-C(P-1) = PC-P+2-CP+C =$ $\mathbf{C-P+2}$.

A · PAST ESA QUESTIONS, FULLY SOLVED

Q5.1 Derive the phase rule for a system of C components and P phases in equilibrium.

Jun-24 · UE23 · Set 1 · 4 marks

1. See derivation Q5-D.

ANSWER $F = C - P + 2$

Q5.2 Derive the phase rule for a system of C components and P phases in equilibrium.

Jan-25 · UE24 · Set 1 · 4 marks

1. See derivation Q5-D.

ANSWER $F = C - P + 2$

Q5.3 (i) Define component. Give the number of phases and components for the closed system $MgCO_3(s) \rightleftharpoons MgO(s) + CO_2(g)$. (ii) Define eutectoid and peritectic points in the Fe–C system.

Jan · UE23A · Set 2 · 4 marks

1. (i) **Component:** the smallest number of independently variable constituents by which the composition of every phase can be expressed. Phases: $MgCO_3(s)$, $MgO(s)$, $CO_2(g)$: $P = 3$. Species 3 − 1 reaction = **$C = 2$** (e.g. choose MgO and CO_2: $MgCO_3 = MgO + CO_2$).
2. (ii) Eutectoid: one solid phase transforms isothermally into two other solids (austenite → ferrite + cementite, 727 °C, 0.76 wt% C). Peritectic: liquid + one solid → a new solid (liquid + δ-ferrite → austenite, 1493 °C, 0.17 wt% C). **Flag: Fe–C is not in the current material; values are standard textbook figures.**

ANSWER $P = 3, C = 2$

Q5.4 (i) Why is the phase rule $F = C - P + 2$ for water but $F = C - P + 1$ for the Pb–Ag system? (ii) For $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ in a closed container give the number of components and phases.

Jan-25 · UE24 · Set 2 · 6 marks

1. (i) Water is a one-component system with T and P both variable. For Pb–Ag, $C = 2$ so $F = 4 - P$, giving $F = 3$ for one phase — a 3-D diagram. Pressure is therefore fixed (≈1 atm) and, since solid–liquid equilibria (condensed systems, no gas) hardly depend on pressure, the rule reduces to $F = C - P + 1$.
2. (ii) $P = 3$ ($CaCO_3$, CaO, CO_2); species 3 − 1 reaction ⇒ $C = 2$.

ANSWER **(ii) $P = 3, C = 2$**

Q5.5 (i) Draw a neat labelled phase diagram of water (areas, lines, triple point, critical point, pressure, temperature) and calculate the degree of freedom along the freezing-point curve. (ii) What is the condensed phase rule and why is it applied to the two-component Pb–Ag system?

Jan-25 · UE24 · Set 1 · 6 marks

1. (i) Diagram: **Fig. G22.** Triple point O: 0.0098 °C, 4.58 mmHg; critical point A: 374 °C, 218 atm. Along a curve $C = 1, P = 2$: $F = 1 - 2 + 2 = \mathbf{1}$ (univariant).
2. (ii) $F = C - P + 1$: used when pressure is constant and the gas phase is absent (condensed systems), as in Pb–Ag where P has little effect.

ANSWER **$F = 1$ on the freezing curve**

Q5.6 (i) Draw a neat labelled phase diagram of the water system with areas, lines and T, P values at the triple and critical points. (ii) Which phases are in equilibrium on the metastable curve? (iii) Calculate F at the triple point.

Jun-24 · UE23 · Set 2 · 6 marks

1. (i) **Fig. G22.**
2. (ii) Supercooled liquid water ⇌ vapour (curve OA′, an extension of OA below the triple point).
3. (iii) $C = 1, P = 3$: $F = 1 - 3 + 2 = \mathbf{0}$ — invariant.

ANSWER **(ii) liquid + vapour; (iii) F = 0**

Q5.7 (i) Draw the phase diagram of water, label areas and boundaries, and calculate F in the areas, along phase boundaries and at the triple point. (ii) Using the Clausius–Clapeyron equation, discuss why the melting-point curve has a negative slope. (iii) What is the reduced phase rule and when is it used?

Dec-24 · UE23 · 8 marks

1. (i) **Fig. G22**: areas $F = 3 - 1 = 2$; lines $F = 3 - 2 = 1$; triple point $F = 3 - 3 = 0$.
2. (ii) $\frac{dP}{dT} = \frac{\Delta H_{fus}}{T\Delta V}$. Fusion: $\Delta H > 0$; ice → water **decreases** volume ($\Delta V < 0$), so $dP/dT < 0$: raising pressure lowers the melting point. The slope is steep because $|\Delta V|$ is small. (OA and OB have $\Delta V > 0$ and large, so gentle positive slopes.)
3. (iii) $F = C - P + 1$ — for condensed systems at constant pressure.

ANSWER **F = 2, 1, 0; dP/dT < 0 because ΔV < 0**

Q5.8 (i) Draw a neat labelled phase diagram of the lead–silver system indicating areas and lines. Give the temperature and composition at the eutectic point. (ii) For the Fe–C system mention the phases in equilibrium, temperature and wt% C at the peritectic point.

Jun-24 · UE23 · Set 1 · 6 marks

1. (i) **Fig. G23**: Pb melts at 327 °C, Ag at 961 °C; curves AO and BO fall to the eutectic point O at **303 °C, 2.6 % Ag (97.4 % Pb)**; areas: liquid melt; liquid + solid Ag; liquid + solid Pb; solid Pb + solid Ag. At O: $C = 2, P = 3, F = 0$.
2. (ii) Peritectic: liquid + δ-ferrite ⇌ austenite at **1493 °C, 0.17 wt% C** (flag: Fe–C not in current material).

ANSWER **Eutectic: 303 °C, 2.6 % Ag**

Q5.9 (i) Draw a neat labelled phase diagram of the CO_2 system with areas, lines, triple and critical points (T, P). (ii) What is the reduced phase rule? Why is the general Gibbs rule not applied to a two-component system?

Jan · UE23A · Set 2 · 6 marks · NOT ON CURRENT SLIDES

1. (i) Like water but the solid–liquid line leans **right** (positive slope, since solid CO_2 is denser). Triple point **−56.6 °C, 5.11 atm**; critical point **31.1 °C, 73 atm**; at 1 atm CO_2 sublimes (−78.5 °C). **Flag: not in current material; standard values.**
2. (ii) For $C = 2$, $F = 4 - P$ gives $F = 3$ when $P = 1$, needing three axes (T, P, composition). So pressure is fixed: $F = C - P + 1$.

ANSWER **Triple −56.6 °C / 5.11 atm; critical 31.1 °C / 73 atm**

B · PRACTICE QUESTIONS (ANSWERS IN THE KEY AT THE END OF THIS CHAPTER)

Set 1 · Count P, C and F

The exam asks this for $CaCO_3$ and $MgCO_3$. The twist is usually NH_4Cl or an added reactant.

Q5.P1.1 For the system: Ice ⇌ water ⇌ water vapour at the triple point, find P, C and F using $F = C - P + 2$.

Q5.P1.2 For the system: Liquid water in equilibrium with its vapour, find P, C and F using $F = C - P + 2$.

Q5.P1.3 For the system: $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ (closed), find P, C and F using $F = C - P + 2$.

Q5.P1.4 For the system: $MgCO_3(s) \rightleftharpoons MgO(s) + CO_2(g)$ (closed), find P, C and F using $F = C - P + 2$.

Q5.P1.5 For the system: $NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$, starting from NH_4Cl only, find P, C and F using $F = C - P + 2$.

Q5.P1.6 For the system: $NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$ with excess NH_3 added, find P, C and F using $F = C - P + 2$.

Q5.P1.7 For the system: Saturated NaCl solution + excess solid NaCl + vapour, find P, C and F using $F = C - P + 2$.

Q5.P1.8 For the system: Two immiscible liquids (oil and water) and their vapour, find P, C and F using $F = C - P + 2$.

Q5.P1.9 For the system: Rhombic sulphur $\rightleftharpoons$ monoclinic sulphur $\rightleftharpoons$ sulphur vapour, find P, C and F using $F = C - P + 2$.

Q5.P1.10 For the system: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ in a closed vessel, starting from PCl_5 only, find P, C and F using $F = C - P + 2$.

Q5.P1.11 For the system: $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, all three initially present, find P, C and F using $F = C - P + 2$.

Q5.P1.12 For the system: $Na_2SO_4 \cdot 10H_2O(s) \rightleftharpoons Na_2SO_4(s) + 10\ H_2O(g)$, find P, C and F using $F = C - P + 2$.

Set 2 · Number of variables and constraints (derivation check)

Using the derivation to confirm F = C − P + 2.

Q5.P2.1 For a system of $C = 2$ components in $P = 1$ phases: how many composition variables, total variables, constraints and what is F?

Q5.P2.2 For a system of $C = 2$ components in $P = 2$ phases: how many composition variables, total variables, constraints and what is F?

Q5.P2.3 For a system of $C = 2$ components in $P = 3$ phases: how many composition variables, total variables, constraints and what is F?

Q5.P2.4 For a system of $C = 3$ components in $P = 2$ phases: how many composition variables, total variables, constraints and what is F?

Q5.P2.5 For a system of $C = 3$ components in $P = 4$ phases: how many composition variables, total variables, constraints and what is F?

Set 3 · Water diagram

Draw Fig. G22 from memory, then check.

Q5.P3.1 What are F in an area, on a line, and at the triple point of water? Name the systems.

Q5.P3.2 Name the curves OA, OB, OC and OA′ and the phases in equilibrium on each.

Q5.P3.3 Give the temperature and pressure at the triple point and critical point.

Q5.P3.4 Why does water stop existing as a liquid above the critical point?

Q5.P3.5 What happens to the melting point of ice if pressure increases, and why?

Q5.P3.6 State the F-value and meaning of: ice + water at 1 atm. What is the temperature?

Set 4 · Clausius–Clapeyron slopes (extension)

Not asked numerically in past papers — included to make the sign argument concrete (Added for clarity).

Q5.P4.1 Ice melting at 273.15 K: $\Delta H_{fus} = 6010\ \text{J mol}^{-1}$, $\Delta V = -1.63 \times 10^{-6}\ \text{m}^3\ \text{mol}^{-1}$. Find dP/dT.

Q5.P4.2 Water boiling at 373.15 K: $\Delta H_{vap} = 40700\ \text{J mol}^{-1}$, $\Delta V \approx 30.6 \times 10^{-3}\ \text{m}^3\ \text{mol}^{-1}$. Find dP/dT and compare with Set 4 Q1.

Q5.P4.3 Predict the sign of the slope for a substance that expands on melting.

Set 5 · Reduced phase rule and the Pb–Ag system

Q5.P5.1 Why do we use $F = C - P + 1$ for Pb–Ag?

Q5.P5.2 Find F at the eutectic point of Pb–Ag and name the three phases present.

Q5.P5.3 Find F along the freezing-point curves AO and BO and in the one-phase liquid region.

Q5.P5.4 A Pb–Ag melt contains 10 % Ag. It is cooled slowly from 700 °C. Describe the cooling path.

Q5.P5.5 What is a eutectic mixture? Give two properties.

Q5.P5.6 How does the phase rule explain why the cooling curve of the eutectic composition shows a single halt?

Set 6 · Cooling curves (thermal analysis)

Q5.P6.1 Sketch the cooling curve of a pure metal and explain the flat portion.

Q5.P6.2 Sketch the cooling curve of a binary mixture (not eutectic) and explain the break and the halt.

Q5.P6.3 How is a phase diagram constructed from cooling curves?

C · ANSWER KEY

Set 1 · Count P, C and F

Q5.P1.1 1. single substance, three physical phases.

ANSWER $P = 3, C = 1, F = 1 - 3 + 2 = \mathbf{0}$ **(invariant)**

Q5.P1.2 1. one substance.

ANSWER $P = 2, C = 1, F = 1 - 2 + 2 = \mathbf{1}$ **(univariant)**

Q5.P1.3 1. 3 species − 1 reaction.

ANSWER $P = 3, C = 2, F = 2 - 3 + 2 = \mathbf{1}$ **(univariant)**

Q5.P1.4 1. 3 species − 1 reaction.

ANSWER $P = 3, C = 2, F = 2 - 3 + 2 = \mathbf{1}$ **(univariant)**

Q5.P1.5 1. gas has NH_3 : HCl = 1 : 1, an extra restriction.

ANSWER $P = 2, C = 1, F = 1 - 2 + 2 = \mathbf{1}$ **(univariant)**

Q5.P1.6 1. the 1 : 1 restriction is lost.

ANSWER $P = 2, C = 2, F = 2 - 2 + 2 = \mathbf{2}$ **(bivariant)**

Q5.P1.7 1. NaCl and water; solid, solution, vapour.

ANSWER $P = 3, C = 2, F = 2 - 3 + 2 = \mathbf{1}$ **(univariant)**

Q5.P1.8 1. two components; liquid, liquid, vapour.

ANSWER $P = 3, C = 2, F = 2 - 3 + 2 = \mathbf{1}$ **(univariant)**

Q5.P1.9 1. one component, three phases.

ANSWER $P = 3, C = 1, F = 1 - 3 + 2 = \mathbf{0}$ **(invariant)**

Q5.P1.10 1. single gas phase; composition fixed by $[PCl_3] = [Cl_2]$.

ANSWER $P = 1, C = 1, F = 1 - 1 + 2 = \mathbf{2}$ **(bivariant)**

Q5.P1.11 1. 3 species − 1 reaction.

ANSWER $P = 1, C = 2, F = 2 - 1 + 2 = \mathbf{3}$ **(multivariant)**

Q5.P1.12 1. salt and water as components; hydrate, anhydrate, vapour.

ANSWER $P = 3, C = 2, F = 2 - 3 + 2 = \mathbf{1}$ **(univariant)**

Set 2 · Number of variables and constraints (derivation check)

Q5.P2.1 1. Variables $= P(C-1) + 2$; constraints $= C(P-1)$.

ANSWER **Composition: 1; total variables: 3; constraints: 0;** $F = 3 - 0 = \mathbf{3}$

Q5.P2.2 1. Variables $= P(C-1) + 2$; constraints $= C(P-1)$.

ANSWER **Composition: 2; total variables: 4; constraints: 2;** $F = 4 - 2 = \mathbf{2}$

Q5.P2.3 1. Variables $= P(C-1) + 2$; constraints $= C(P-1)$.

ANSWER **Composition: 3; total variables: 5; constraints: 4;** $F = 5 - 4 = \mathbf{1}$

Q5.P2.4 1. Variables $= P(C-1) + 2$; constraints $= C(P-1)$.

ANSWER **Composition: 4; total variables: 6; constraints: 3;** $F = 6 - 3 = \mathbf{3}$

Q5.P2.5 1. Variables $= P(C-1) + 2$; constraints $= C(P-1)$.

ANSWER **Composition: 8; total variables: 10; constraints: 9;** $F = 10 - 9 = \mathbf{1}$

Set 3 · Water diagram

Q5.P3.1 1. $F = 3 - P$.

ANSWER **Area** $F = 2$ **(bivariant); line** $F = 1$ **(univariant); triple point** $F = 0$ **(invariant).**

Q5.P3.2

ANSWER **OA vaporisation (water ⇌ vapour); OB sublimation (ice ⇌ vapour); OC fusion (ice ⇌ water, negative slope); OA′ metastable supercooled water ⇌ vapour.**

Q5.P3.3

ANSWER **Triple point 0.0098 °C, 4.58 mmHg; critical point 374 °C, 218 atm.**

Q5.P3.4

ANSWER **The liquid–vapour interface vanishes — above the critical temperature no pressure can liquefy the vapour.**

Q5.P3.5

ANSWER **It decreases: ice melts to a smaller volume ($\Delta V < 0$), so by Clausius–Clapeyron $dP/dT < 0$.**

Q5.P3.6

ANSWER **$P = 2$, $F = 1$; at fixed 1 atm T is fixed at 0 °C (normal melting point).**

Set 4 · Clausius–Clapeyron slopes (extension)

Q5.P4.1

1. $dP/dT = \Delta H/(T\Delta V)$.

ANSWER **$dP/dT = -1.35 \times 10^{7}$ Pa K^{-1} ≈ -133 atm K^{-1} — large and negative: very steep curve OC.**

Q5.P4.2

1. Same equation.

ANSWER **$dP/dT \approx 3567$ Pa K^{-1} ≈ 35.2×10^{-3} atm K^{-1} — positive and tiny by comparison (large ΔV), so OA is gentle.**

Q5.P4.3

ANSWER **$\Delta V > 0 \Rightarrow dP/dT > 0$: melting curve leans right (e.g. CO_2).**

Set 5 · Reduced phase rule and the Pb–Ag system

Q5.P5.1

1. Condensed system (no gas), pressure fixed.

ANSWER **Because $C = 2$ would need 3-D at $F = C - P + 2$; at constant pressure one variable is removed.**

Q5.P5.2

1. $C = 2, P = 3$.

ANSWER **$F = 2 - 3 + 1 = 0$ (invariant). Phases: solid Pb, solid Ag and liquid melt (303 °C, 2.6 % Ag).**

Q5.P5.3

ANSWER **Curves: $P = 2$ so $F = 1$ (univariant); liquid region $P = 1$ so $F = 2$ (bivariant).**

Q5.P5.4

1. 10 % Ag is richer in Ag than the eutectic (2.6 % Ag), so the melt cools until it meets the Ag freezing-point curve (O to the Ag melting point): solid Ag separates first and the melt becomes richer in Pb.

ANSWER **Solid Ag separates first on the Ag-side curve; the melt composition slides down that curve to O (303 °C, 2.6 % Ag); there Pb and Ag crystallise together at constant temperature (halt, F = 0); below 303 °C: solid Pb + solid Ag. (A melt with less than 2.6 % Ag would separate Pb first.)**

Q5.P5.5

ANSWER **A mixture of two components with the lowest freezing point of all compositions; it has a definite composition and a sharp melting point and solidifies at constant temperature (F = 0).**

Q5.P5.6

ANSWER **At the eutectic $F = 0$: temperature stays fixed while all three phases coexist until solidification is complete.**

Set 6 · Cooling curves (thermal analysis)

Q5.P6.1

ANSWER **Smooth fall, then a halt at the freezing point (solid and liquid coexist, $F = C - P + 1 = 1 - 2 + 1 = 0$), then a fall again. (Fig. G24)**

Q5.P6.2

ANSWER **Fall → break (change of slope) when the first solid appears (heat of crystallisation slows cooling, $F = 1$) → fall → halt at the eutectic temperature ($F = 0$) → fall.**

Q5.P6.3

ANSWER **Record cooling curves for several compositions; read the break and halt temperatures; plot them against composition and join the break points (liquidus) and the common halt temperature (eutectic line).**

CHAPTER 6 · TOPICS T13

Computational chemistry

1 solved past questions | 10 practice questions | 0 theory answers

A recently added topic: one question in 2025 so far (definition + three applications). Expect short-answer questions.

RECAP · FORMULAS YOU NEED

Classical: MM, MD | Quantum: semi-empirical, a

A · PAST ESA QUESTIONS, FULLY SOLVED

Q6.1 (i) What is computational chemistry? Give any three applications of computational chemistry.

Jan-25 · UE24 · Set 2 | 7 marks

1. **Definition:** a set of techniques that solves chemical problems by simulating molecular, biological or materials systems to give reliable, accurate, atomic-level information, using theoretical chemistry built into efficient computer programs.
2. **Three applications:** (1) predicting molecular properties — HOMO/LUMO and UV-Vis, IR, NMR spectra; (2) studying reaction mechanisms and kinetics; (3) designing new materials (conductivity, thermal stability); (also drug discovery by simulating drug–target interaction).

ANSWER **Definition + any three applications**

B · PRACTICE QUESTIONS (ANSWERS IN THE KEY AT THE END OF THIS CHAPTER)

Set 1 · Short answers

Practise writing 2–3-mark answers in your own words.

Q6.P1.1 Define computational chemistry and name the first theoretical investigation.

Q6.P1.2 List four areas in which computational modelling is used.

Q6.P1.3 Differentiate classical and quantum methods of computational chemistry.

Q6.P1.4 What is molecular mechanics? Why is it fast?

Q6.P1.5 How does DFT differ from ab initio methods?

Q6.P1.6 Rank MM, semi-empirical, DFT and ab initio by speed and accuracy.

Q6.P1.7 List three advantages of computational chemistry. Does it replace experiments?

Q6.P1.8 Write the steps of the electronic-structure determination of ethylene.

Q6.P1.9 Describe the bonding and HOMO/LUMO of ethylene.

Q6.P1.10 Give two scopes and two advantages of AI-enhanced computational chemistry.

C · ANSWER KEY

Set 1 · Short answers

Q6.P1.1 — ANSWER: Techniques to simulate chemical systems and compute properties at the atomic level. First theoretical work: Heitler and London, 1927.

Q6.P1.2 — ANSWER: Molecular properties/spectra, reaction mechanism and kinetics, material design, drug discovery.

Q6.P1.3 — ANSWER: Classical (molecular mechanics, molecular dynamics): Newtonian, atoms as spheres and bonds as springs, electrons ignored, very fast. Quantum (semi-empirical, ab initio, DFT): based on the Schrödinger equation, treat electrons explicitly, slower but more accurate.

Q6.P1.4 — ANSWER: Model of a molecule as balls (atoms) joined by springs (bonds), energy from classical force fields; large molecules such as steroids can be optimised in seconds.

Q6.P1.5 — ANSWER: DFT derives the electron density directly instead of a conventional many-electron wave function; faster than ab initio but slower than semi-empirical.

Q6.P1.6 — ANSWER: Speed: MM > semi-empirical > DFT > ab initio. Accuracy (generally): ab initio > DFT > semi-empirical > MM.

Q6.P1.7 — ANSWER: Easier, cheaper and safer than experiments. It does not replace experiments; it helps chemists design and develop desired products.

Q6.P1.8 — ANSWER: 1 Define geometry of C_2H_4; 2 select a basis set; 3 apply the method to get orbital energies and density; 4 solve the equations for the wave function / electron density; 5 visualise the orbitals (HOMO = π, LUMO = π*).

Q6.P1.9 — ANSWER: Each carbon is sp^2: 2 C–H σ bonds + 1 C–C σ bond; one unhybridised p orbital per carbon forms the π bond. HOMO = π, LUMO = π*.

Q6.P1.10 — ANSWER: Scopes: accelerated molecular discovery; property prediction; reaction-pathway mapping; structure generation. Advantages: massive speed-up (hours/days → milliseconds once trained); large-scale screening; pattern recognition of structure–property relations.